

Strip cropping designed for maintaining productivity increases ground beetle biodiversity

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This study shows that strip cropping -- planting different crops in strips on the same field -- enhances the taxonomic diversity of ground beetles relative to corresponding monocultures in multiple experiments with different crops in the Netherlands. While these findings are **important** for demonstrating the potential beneficial effects of this form of intercropping, the information presented is **incomplete** with regard to sampling design and data obtained.

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Abstract

Global biodiversity is declining at an unprecedented rate, with agriculture as a major driver. There is mounting evidence that intercropping can increase insect biodiversity while maintaining or increasing yield. Yet, intercropping is often considered impractical for mechanized farming systems. Strip cropping is pioneered by Dutch farmers as it is compatible with standard farm machinery. Here, we use ground beetle data from four experimental locations across four years of measurements because these are good agrobiodiversity indicators and hold keystone positions in agroecosystem foodwebs. We show that strip cropping systems that are designed for retaining productivity can also enhance ground beetle biodiversity, without incurring major yield loss. Strip cropped fields had on average 15% more ground beetle species and 30% more individuals than monocultural fields. The increase in field-level beetle species richness in organic agriculture through strip cropping approached increases found for other readily deployed biodiversity conservation methods, like shifting from conventional to organic agriculture (+19% - +23%). This makes strip cropping a useful tool for bending the curve of biodiversity loss without compromising food production.

Introduction

Insects account for 80% of the animal species in the world and, therefore, the recently reported unprecedented rate of insect decline is cause for alarm about the state of biodiversity on Earth (Dirzo *et al.*, 2014; Hallmann *et al.*, 2017; van Klink *et al.*, 2020). An array of biodiversity metrics provide strong evidence for declines of especially terrestrial insects across all continents (van Klink *et al.*, 2020). This includes declines in abundance of both common and rare species (Hallmann *et al.*, 2017; Seibold *et al.*, 2019; Pilotto *et al.*, 2020; van Klink *et al.*, 2023) and in total species richness and changes in assembly composition (Wagner *et al.*, 2021; Blowes *et al.*, 2022). In some areas, terrestrial insect biomass decreased by 75% in the last 30 years (Hallmann *et al.*, 2017), and a further 40% of all species is estimated to face extinction by 2100 (IPBES, 2019). Insects are essential for crop production through their role in decomposition, pollination, pest control and sustaining food webs. Therefore, erosion of insect communities can have potentially devastating effects on ecosystem functioning, provision of ecosystem services and ultimately on human civilization (IPBES, 2019). The main drivers of insect biodiversity decline are habitat loss due to conversion to agriculture, pollution, invasive species and climate change (Díaz *et al.*, 2019; Wagner *et al.*, 2021; Müller *et al.*, 2023).

Strategies for biodiversity conservation in conjunction with adequate food production require understanding of how biodiversity responds to agricultural management (Saunders, 2020). The idea of land sparing seeks to conserve natural habitats by separating high-yielding agriculture from protected natural habitats, while land sharing combines food production with biodiversity conservation on agricultural land (Green *et al.*, 2005; Loconto *et al.*, 2020). Combining both pathways is vital (Fischer *et al.*, 2008; Scariot, 2013) as land sparing counteracts land-use change and land sharing makes agricultural landscapes better suited to support biodiversity and associated ecosystem services that support crop production (Dainese *et al.*, 2019). Ideally, sustainable agricultural practices retain yield and enhance biodiversity, preventing the need to convert natural habitats to agricultural land to maintain food production (Tscharntke *et al.*, 2021).

Increasing crop heterogeneity can facilitate biodiversity conservation in highly productive agricultural landscapes without compromising yield (Martin-Guay *et al.*, 2018; Sirami *et al.*, 2019). Crop diversification can enhance niche complementarity by creating heterogeneous habitats and increasing availability and diversity of resources (Lichtenberg *et al.*, 2017; Tamburini *et al.*, 2020). A promising crop diversification strategy is strip cropping, where crops are grown in alternating strips, wide enough for using standard agricultural machines yet narrow enough to facilitate ecological interactions among crops (Ditzler *et al.*, 2021). Crops that are grown in strips may benefit from increased resource use efficiency, and the suppression of pests and diseases (Tajmiri *et al.*, 2017b; Ditzler *et al.*, 2021; Juventia *et al.*, 2021; Alarcón-Segura *et al.*, 2022; Cuperus *et al.*, 2023; Karssemeijer *et al.*, 2023; Rakotomalala *et al.*, 2023; Croijmans *et al.*, 2024) without major yield compromises (Tajmiri *et al.*, 2017a, b; Mollaei *et al.*, 2020; van Oort *et al.*, 2020; Juventia *et al.*, 2021; Campanelli *et al.*, 2023). Growing multiple crops on a field may foster a larger diversity of organisms than monocultures through greater plant species richness that cascades into richer herbivore and predator communities (Crutsinger *et al.*, 2006; Cuperus *et al.*, 2024). Moreover, the expected increase in available and potentially complementary niches within the agricultural field due to higher spatial diversity in crops can result in admixture of communities related to individual crops (Hummel *et al.*, 2012), or the creation of completely new communities by enhanced richness of agriculture-related species, and/or the occurrence of species rarely found in agricultural fields (Rischen *et al.*, 2021). So far, it is not well understood if and how insect communities respond to strip cropping across distinct crops.

Here, we present data of four years of pitfall trapping of ground beetles at several moments during the growing season in 14 crops at four organic experimental farms across the Netherlands where strip cropping was compared to monocultures. Ground beetle community dynamics are sensitive to changes in farming practices and are frequently used to examine agricultural sustainability (Holland *et al.*, 2002 [↗](#); Makwela *et al.*, 2023 [↗](#)). Furthermore, ground beetle species are important for maintaining ecological functions as they comprise scavengers and predators of (weed) seeds, detritivores (e.g., collembolas and earthworms) and herbivores (e.g., aphids and caterpillars). As such, they are often used as indicator group for wider insect diversity in agroecosystems (Gerlach *et al.*, 2013 [↗](#)). We first examine whether strip cropping fields have greater ground beetle richness and activity density than monocultural fields. Next, we present biodiversity metric changes for individual crops to assess the benefits of strip cropping for ground beetle biodiversity. We then evaluate whether ground beetle community changes are caused by admixture of communities, whether these assemblages promote species associated with agriculture or with natural ecosystems, and whether they contain rare species.

Results

A total of 49,199 ground beetles belonging to 71 species were caught using pitfall traps over four years at four different organically managed experimental locations in The Netherlands: 40,159 at Almere; 3,777 at Lelystad; 1,126 at Valthermond; and 4,137 at Wageningen (Table S1 [↗](#), S2 [↗](#)).

Strip cropping enhances ground beetle richness

Strip cropping fields had on average 15% higher ground beetle taxonomic richness than monoculture fields after rarefaction to the number of samples of the least-sampled crop configuration ($\beta_0 = 0.151$, SE = 0.044, $p < 0.001$; Fig. 1c,d [↗](#)). However, strip cropping fields did not harbour more species than monocultural fields with the highest ground beetle richness ($\beta_0 = -0.008$, SE = 0.037, $p = 0.821$; Fig. 1d [↗](#)). The difference in field-level taxonomic richness could not be explained by an increase in the number of ground beetle species per crop in strip cropping compared to monocultures. At crop-level, the 5% increase in ground beetle taxonomic richness in strip cropping was not statistically significant (Fig. 1f [↗](#)). Similarly, crop-level absolute evenness, inverse Simpson index and Shannon entropy did not differ significantly among crop configurations (Fig. 1h-j [↗](#), S1c,d [↗](#), S2 [↗](#)). The effect of crop configuration on crop-level taxonomic richness was variable and was not associated with location or crop species. The effect ranged from 56% more species in potato in monoculture at Wageningen in 2022, to 136% more species in barley in strip cropping at Wageningen in 2020 (Fig. 1k [↗](#)).

Strip cropping enhances ground beetle activity density

Ground beetle activity density was on average 30% higher in strip cropping fields than in monoculture fields ($\beta_0 = 0.303$, SE = 0.121, $p = 0.012$; Fig. 1e [↗](#)), based on rarefaction. However, there was no significant difference in activity density between the strip cropping fields and monocultures with crops that harboured the richest beetle communities ($\beta_0 = -0.110$, SE = 0.088, $p = 0.215$). Crop-level activity density of ground beetles was not affected by crop configuration (Fig. 1g [↗](#), S1a,b [↗](#)).

Crop configuration alters ground beetle community composition

Ground beetle communities were significantly influenced by crop configuration, but the effects of location, year and crop dominated those of crop configuration (Fig. S3 [↗](#)-S5 [↗](#), Table 1 [↗](#), S3 [↗](#)). The context-dependency of configuration effects on ground beetle communities is illustrated in redundancy analyses of ground beetle assemblages per crop combination at Wageningen in 2021 and 2022 (Fig. S6 [↗](#)). Here, we found distinct ground beetle communities among crop

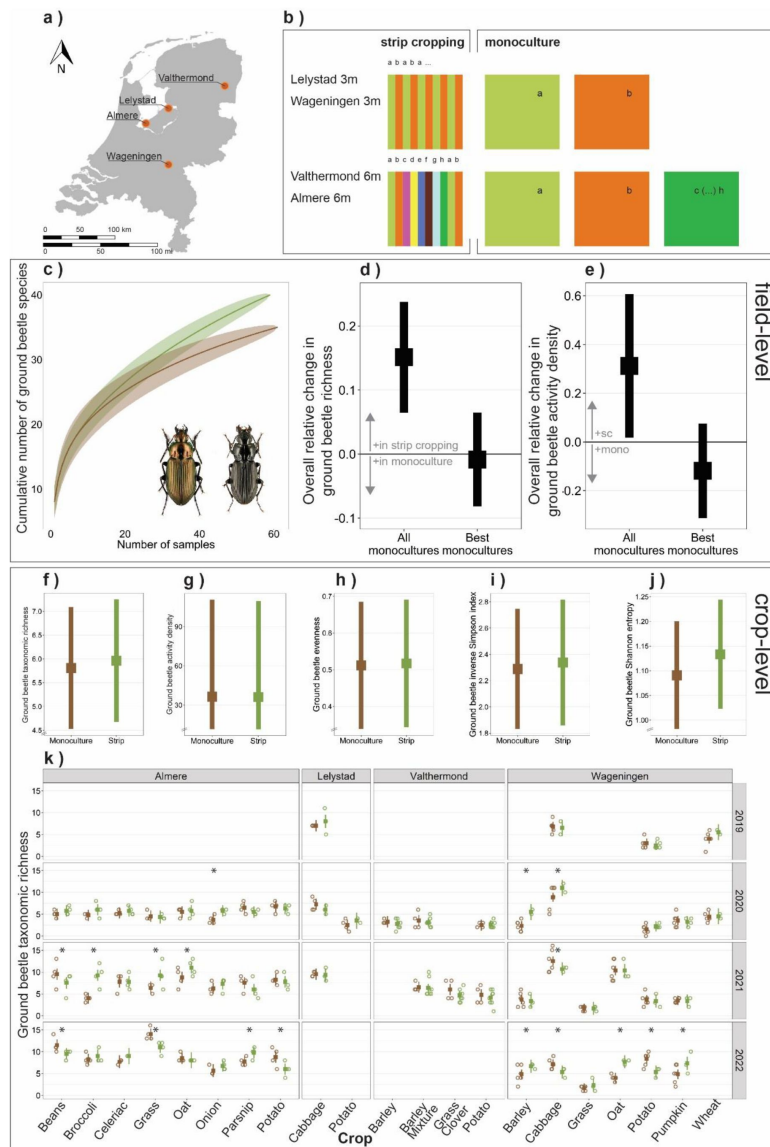


Fig. 1.

Effect of crop configuration (monoculture versus strip cropping) on ground beetle biodiversity.

(a) Location of experimental sites in the Netherlands. (b) Field set-up of the two crop configurations: monoculture and strip cropping. At Lelystad and Wageningen, strip cropping consisted of 3-m wide crop strips of two crops (pairs), and multiple crop pairs were assessed. At Almere and Valthermond, strip cropping consisted of 6-m wide crop strips of eight crops combined (Fig. S7). (c) Sample-based species accumulation curves of all year series from monocultures (brown) and strip cropping (green), in Almere from 2021 and 2022. This was the only location where an equal number of samples were taken in the monocultures and in strip cropping on a similar area. Ground beetle species include *Poecilus cupreus* (left) and *Pterostichus melanarius* (right). Photo credit: Ortwin Bleich, retrieved from: www.eurocarabidae.de (d-e) Overall relative change in field-level ground beetle (d) taxonomic richness and (e) activity density. Positive values indicate higher richness or activity density in strip cropping, negative values in monocultures. (f-j) Overall relative effect of crop configuration on ground beetle (f) taxonomic richness, (g) activity density, (h) absolute evenness, (i) inverse Simpson index, and (j) Shannon entropy. (k) Effect of crop configuration on ground beetle taxonomic richness for each combination of location, year and crop. Barley-mixture consists of a mixture of barley-bean (2020) or barley-pea (2021). Squares indicate estimated means, the bar indicates the 95% confidence interval. Asterisks indicate significant differences among the crop configurations. Empty panels indicate combinations of years and locations that were not sampled. When no estimated mean and confidence interval are shown, crops were not grown or sampled in that year. Open circles indicate individual year series to visualize sample size (Table S7).

configurations for pumpkin, barley and potato in 2021, and for cabbage and oat in 2022. In the other cases the difference between crop configurations was not significant (Fig. S6, Table S4). Moreover, in all crop combinations except for potato-grass in 2021, the difference in ground beetle communities between the monocultures of the constituent crops were significant, while this was never the case for ground beetle communities of crops in strip cropping (Table S4). This indicates that strip cropping leads to overlapping crop-related communities. However, these results could be spatially biased as samples from different crops were in closer proximity of each other in strip cropping than among monocultures (Fig S7d).

Rare and common species indicators for strip cropping

Eleven ground beetle species were significantly associated with crop configuration: four species associated with strip cropping and seven with monocultures (Fig. 2, Table S5). Specifically, the rare *Harpalus griseus* ($p=0.001$, specificity=0.900) and the more common *Anchomenus dorsalis* ($p=0.001$, specificity=0.806) were strongly associated with strip cropping (Fig. 2, Table S5).

Discussion

It is well established that different crop types have distinct ground beetle communities (Holland and Luff, 2000; Eyre *et al.*, 2009) and that increasing habitat diversity by including multiple crops in a field can enhance ground beetle diversity (Puliga *et al.*, 2023; Cuperus *et al.*, 2024). Our study shows that strip cropping increased field-level ground beetle richness by 15%. However, ground beetle richness in crops grown in strip cropping and in monocultures were mostly similar, and there were relatively few ground beetle species that were unique in strip cropping. In fact, our indicator species analysis showed that more species were associated with monocultures than with strip cropping. This indicates that the 15% increase in richness at the field level can be mostly attributed by the higher number of crops in strip cropped fields that harboured crop-specific ground beetle communities, and that there was only a limited mixture of ground beetle species among crops in strip crops. This is in line with findings from Anderson *et al.* (2024) who found that ground beetle movement is reduced by crop edges. Further research on movement behaviours of ground beetles at crop edges might help explain how ground beetles distribute themselves within a strip cropping field, and whether they utilize the different resources provided by a more diverse cropping system.

The 30% higher ground beetle activity density in strip cropping fields compared to monoculture fields may be explained by a more stable and diverse habitat with refuges and alternative resources in strip crops (Ratnadass *et al.*, 2012). Crop diversification enhances prey biomass for ground beetles (Lichtenberg *et al.*, 2017) and increases weed seed biomass as compared to monocultures (Carbonne *et al.*, 2022; Ditzler *et al.*, 2023), both of which reduce bottom-up control by increased food provision. In addition, insectivorous farmland birds regularly prefer more diverse fields for foraging, consequently increasing top-down control (Josefsson *et al.*, 2017). Crop diversification might thus influence bottom-up and top-down control of ground beetles in opposing ways and the relative strength of each might explain the variation in responses of ground beetle activity density among locations and crops. Multi-taxa evaluation of future strip cropping studies will be a valuable approach to increase understanding of biomass flows and trophic interactions within agricultural systems.

The ground beetles *H. griseus* and *A. dorsalis* were specifically associated with strip cropping. The granivorous *H. griseus* may have benefitted from the relatively high weed abundance and diversity in the strip crops (Carbonne *et al.*, 2022; Ditzler *et al.*, 2023). *A. dorsalis* particularly associated with strip cropping at Almere, which could have been caused by the closeness of preferred hibernation locations such as grass-clover strips in the strip cropping configuration,

Location	Predictor	Df	Sum sq	R2	F	P
All locations	Crop configuration	1	1.68	0.01	7.32	0.001
	Location	2	47.7	0.29	103.8	0.001
	Location : Year	6	10.7	0.06	7.72	0.001
	Location : Year : Crop species	31	33.3	0.20	4.67	0.001
	Crop configuration : Location	2	0.63	0.00	1.37	0.149
	Crop configuration : Location : Year	6	2.31	0.01	1.67	0.012
	Crop configuration : Location : Year : Crop species	31	14.8	0.09	2.08	0.001
	<i>Residual</i>	<i>245</i>	<i>56.3</i>	<i>0.34</i>		
	<i>Total</i>	<i>324</i>	<i>167.4</i>	<i>1.00</i>		

Table 1.

Effect of crop configuration on ground beetle community composition.

Results from permanova analysis using Hellinger's transformation for data from the three locations with species level data (see **Table S4** [↗](#) for analyses per location). Crops were a nested variable within years, as these differed among years. Years were nested in locations, as the years that were studied differed among locations. Bold letters indicate significant effects ($\alpha = 0.05$).

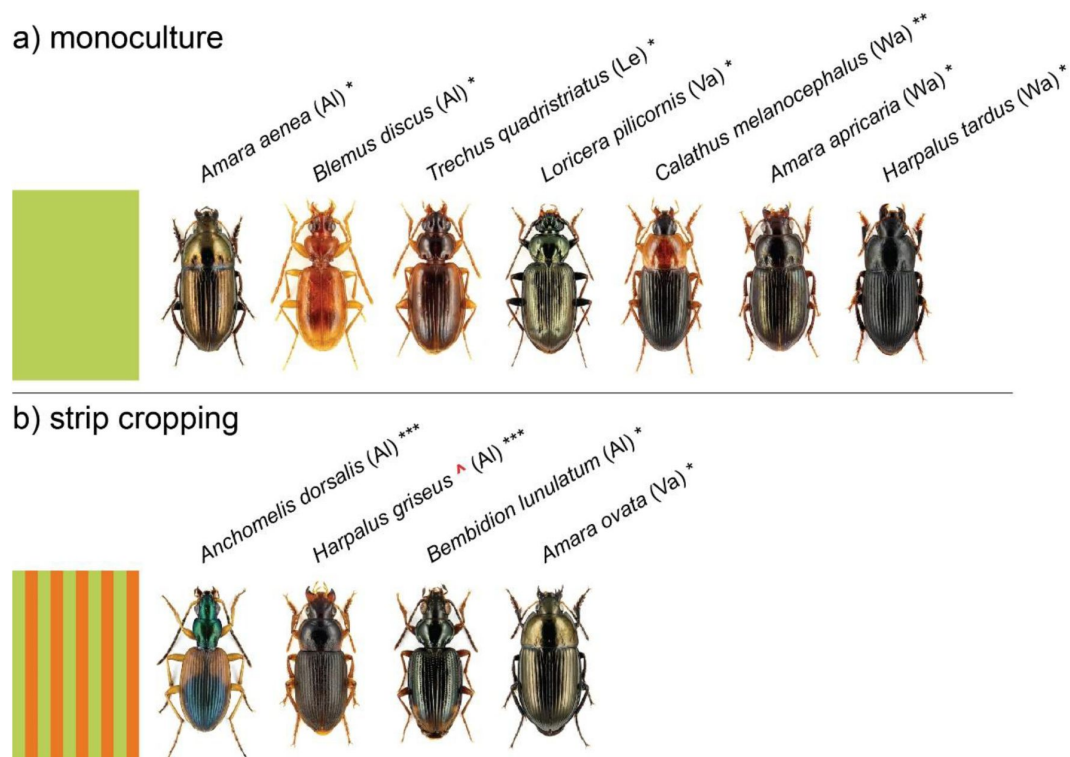


Fig. 2.

Ground beetle species associated with crop configuration (monoculture (a) versus strip cropping (b)).

Results obtained by indicator species analyses with the four locations analysed separately (Table S6). Locations are indicated between brackets (Al=Almere; Le = Lelystad; Va = Valthermond; Wa = Wageningen). Data from Almere in 2020 was excluded from the analysis as ground beetles were identified up to genus level only. An asterisk indicates the significance of the association between ground beetle species and crop configuration (*: $\alpha < 0.05$; **: $\alpha < 0.01$; ***: $\alpha < 0.001$). Rare species indicated by red "▲" (Turin, 2000). Photo credit: Ortwin Bleich, retrieved from: www.eurocarabidae.de.

compared to monoculture crops (Eyre *et al.*, 2009; Marrec *et al.*, 2017). Seven out of the eleven crop configuration indicator species in our study are typical eurytopic species, adapted to a wide range of environments and common to the Netherlands (Turin, 1991; Turin, 2000). *Amara aenea*, *Calathus melanocephalus*, *Loricera pilicornis* and *Trechus quadristriatus* were associated with monocultures. These polyphagous, generalist species are strongly adapted to crop monocultures characterised by large-scale physical disturbance through tillage and harvest (Kromp, 1999). Overall, our data indicate that multiple ground beetle species have a preference for a specific crop configuration.

Biodiversity gains in our study did in most cases not coincide with productivity losses (Table S6). Earlier studies on crop yield in the strip cropping fields in Lelystad and Wageningen during the years of pitfall trapping show a yield decrease when strip cropping cabbage (Carrillo-Reche *et al.*, 2023) and wheat (Ditzler *et al.*, 2023), whereas potato yield was unaffected by crop configuration (Ditzler *et al.*, 2023). In Almere bean and parsnip yields were higher in strip cropping, whereas oat and onion yield were lower (Juventia *et al.*, accepted). In Valthermond crop productivity was similar for monocultures and strip cropping (Table S6). Therefore, strip cropping has the potential to reconcile farmland biodiversity conservation and agricultural production.

We show that changing crop configuration from monoculture to strip cropping enhances ground beetle richness by 15% and activity density by 30% within agricultural fields. These results show that strip cropping leads to increases in biodiversity that approach those achieved by shifting from conventional to organic farming practices (+19% richness, Lichtenberg *et al.*, 2017; +23% richness, Gong *et al.*, 2022) and by other in-field diversification measures (+23% richness, Lichtenberg *et al.*, 2017; +24% richness, Beillouin *et al.* (2021)). While organic management or in-field diversification measures generally lead to lower productivity (Gong *et al.*, 2022), strip cropping can maintain crop productivity without taking land out of production (Tajmiri *et al.*, 2017a, b; Mollaei *et al.*, 2020; van Oort *et al.*, 2020; Juventia *et al.*, 2021; Campanelli *et al.*, 2023), although occasional yield reductions have been reported (Carillo-Reche *et al.*, 2023; Ditzler *et al.*, 2023). The biodiversity gain in our study was achieved without additional crop-level diversification strategies designed for biodiversity conservation, such as cover cropping or (flowering) companion plants. This increase in biodiversity stacks on top of already higher biodiversity achieved through organic management (Lichtenberg *et al.*, 2017; Gong *et al.*, 2022). Furthermore, the estimated richness effects of strip cropping, i.e., no change at crop level and 15% increase at field level, are conservative because they only consider the pairwise comparison of biodiversity of two or three crops in a strip cropping configuration to monocultures, whereas the inclusion of more crops within strip cropping could further enhance ground beetle biodiversity. Also, measurements were conducted in relatively young strip cropping fields, which were not older than five years (less than a full rotation). Ground beetle communities might require more time to adapt to the new crop configuration (Rusch *et al.*, 2013). The ground beetle communities in our study were dominated by farmland species and contained only few rare species. Nevertheless, even after only four years, specific species strongly associated with strip cropping were found. For Dutch farming systems, which are dominated by non-flowering crops, increased use of flowering crops or inclusion of semi-natural habitats with flowering plants might be necessary on top of strip cropping to increase biodiversity of groups such as pollinators. Strip cropping in combination with other in-field diversification measures, such as the establishment of more perennial semi-natural habitats (Sirami *et al.*, 2019; Rischen *et al.*, 2021) or uptake at larger spatial extents (Tschardt *et al.*, 2021) could further enrich agricultural landscape diversity and help bending the curve of biodiversity loss.

Materials and methods

Study area

A multi-location study was conducted on four organic farms across the Netherlands (**Fig. 1a** [↗](#)). Three experimental farms were managed by Wageningen University & Research (Lelystad, Valthermond, Wageningen) and one commercial farm was managed by *Exploitatie Reservegronden Flevoland* (ERF B.V.) located in Almere. All four locations contained both strip cropping and monocultural crop fields, but differed in soil type, establishment year of the strip cropping experiment, number of crops grown, length of the crop rotation, number of sampled crops and sampling years, and farm and landscape characteristics such as percentage of on-farm semi-natural habitat (SNH), mean field size, and landscape configuration (**Fig. 1b** [↗](#), **S7** [↗](#)). The locations Almere and Lelystad were located in a homogeneous, open polder landscape characterized by intensive arable crop production and non-crop habitats consisting of grass margins, tree lines and watercourses. Valthermond was located in an open, reclaimed peat landscape with intensive arable crop production characterized by long and narrow fields separated by grassy margins and ditches and limited areas of woody elements. The site at Wageningen was located in a more complex landscape with smaller field sizes, and non-crop habitat consisting of woodlots, hedgerows, tree lines, ditches and farmyards.

Experimental lay-out

At Almere and Valthermond the crops in strip cropping were all grown alongside each other, whereas at Lelystad and Wageningen two alternating crops (crop pairs) were grown alongside each other (**Fig. 1b** [↗](#), **S7** [↗](#)). At each location, strip cropping and monoculture fields were always paired on the same experimental field (**Fig. S7** [↗](#)). At Almere eight different crops were grown in alternating strips of 6 m width, including celeriac (*Apium graveolens* var. *rapaceum*), broccoli (*Brassica oleracea* var. *italic*), oat (*Avena sativa*), onion (*Allium cepa*), parsnip (*Pastinaca sativa*), faba bean (*Vicia faba*), potato (*Solanum tuberosum*) and a mix of ryegrass and white clover referred to as grass-clover (*Lolium perenne*/*Trifolium repens*) (**Fig. S7a** [↗](#)). At Lelystad four different crop pairs were grown in alternating strips of 3 m width, including carrot (*Daucus carota* subsp. *sativus*) and onion, white cabbage (*Brassica oleracea* var. *capitata*) and wheat (*Triticum aestivum*), sugar beet (*Beta vulgaris*) and barley (*Hordeum vulgare*), and potato and ryegrass (**Fig. S7b** [↗](#)). At Valthermond eight different crops were grown in alternating strips of 6 m width, including potato, barley, barley mixed with broad bean (*Vicia faba*) in 2020, barley mixed with pea (*Pisum sativum*) in 2021, sweet corn (*Zea mays* convar. *saccharata* var. *rugosa*), sugar beet, common bean (*Phaseolus vulgaris*), and grass-clover (**Fig. S7c** [↗](#)). At Wageningen three different crop pairs were grown in alternating strips of 3 m width, including white cabbage and wheat (2019 – 2021) or oat (2022), barley and pumpkin (*Cucurbita maxima*, 2020 - 2022) or bare soil due to crop failure (2019), and potato and ryegrass (**Fig. S7d** [↗](#)). The crop combinations and neighbors were selected based on literature, expert knowledge and experience of functionality in terms of expected advantages for yield and pest and disease control. Large scale monoculture plots (0.25 ha to 2.30 ha) served as reference, hereafter referred to as monoculture (**Fig. S7** [↗](#)). At Lelystad, Valthermond and Wageningen not each crop grown in strips was present as monoculture in each year, but only those crops for which a monoculture was present were sampled (**Fig. 1k** [↗](#), **Table S7** [↗](#)). All fields were managed according to organic regulations, yet at each location fertilization and weed management reflected regional practices and were adjusted to local soil conditions. Flower strips were sown within the experimental fields at Almere and Valthermond (**Fig. S7** [↗](#), **Table S8** [↗](#)).

Sampling

The ground beetle community was sampled using pitfall traps in all crops for which both a monoculture and strip cropping field were present, at each location and in multiple rounds per year between March and September. The sampled crops, number of rounds and number of pitfalls differed per year and per location (**Table S7** [↗](#)). Pitfall traps consisted of a plastic cup (8.5 cm diameter) placed in the soil so the top of the cup was level with the soil surface. Pitfalls were filled with water mixed with non-perfumed soap and covered with a black roof (12.5 cm diameter). Pitfall trap type was similar in all years and locations and traps were placed at the same specific location per year. For all analyses, year series were made, in which all ground beetle catches from the same pitfall trap were pooled per year.

Statistical analyses

We used R, version 4.2.2 for all statistical analyses.

Effect of crop configuration on field-level richness and activity density

To analyse the difference in species richness and activity density between monocultures and strip cropping configurations at field level, we used rarefaction of samples within the same field. To rarify to an equal sampling intensity, we calculated the average cumulative number of species or individuals within x year series, where x is the largest number of year series available for the crop configuration comparison. Next, we calculated the relative change due to strip cropping by subtracting the number of species or individuals found in the monoculture field from the number found in the strip cropping field, and then dividing the result by the number of species or individuals in the monoculture. This gave the relative change centered around zero, where negative values indicated higher richness in monocultures and positive values higher richness in strip cropping. We then analysed this data using Generalized Linear Mixed Models (GLMM) with a Gaussian distribution, and assessed whether the intercept deviated significantly from zero. As random variables, we used location and year, with year nested in location. We ran these analyses using a dataset that included all comparisons among monocultural fields and strip cropping fields of all locations. To test whether strip cropping was more beneficial for beetle diversity than the most beetle rich monoculture, we separately analysed a dataset that only included the monocultural fields with the highest taxonomic richness or activity density among the constitutive crops of the strip cropping field. Generalized linear models were run using the `glmmTMB` package and tested for model fit using the `DHARMA` package.

Effect of crop configuration on crop-level biodiversity

To quantify biodiversity we used five variables: (1) activity density, the total number of ground beetles found per year series; (2) taxonomic richness, the total number of species or genera (lowest taxonomic level available) found per year series; (3) the inverse Simpson index, the inverse of the sum of proportions of different species over the total abundance ([Simpson, 1949](#) [↗](#)); (4) absolute evenness, the number of effective species calculated by dividing the inverse Simpson index by the taxonomic richness ([Williams, 1964](#) [↗](#)); and (5) Shannon entropy ([Shannon and Weaver, 1949](#) [↗](#)). We chose absolute evenness as our measure for evenness, as this method removes the richness component from the inverse Simpson index and adheres to all requirements for an evenness index ([Smith and Wilson, 1996](#) [↗](#); [Tuomisto, 2012](#) [↗](#)). We included both the inverse Simpson index and Shannon entropy as the former is more sensitive to changes in evenness and the latter to species richness ([DeJong, 1975](#) [↗](#)).

To analyse the effect of crop configuration on ground beetle activity density, taxonomic richness, evenness, inverse Simpson index and Shannon entropy we used GLMM. We constructed models for each response variable, using data from all four locations. In these models we included crop

configuration (monoculture or strip cropping) as a fixed factor. We included location, year and crop as nested random variables in these models. To quantify and visualize the variation in responses between locations, years and crops, we ran generalized linear models (GLM) with a variable that combined these three variables into one, which was also included as a fixed factor.

Here, we also included the interaction between crop configuration and the combined variable for crop, location and year. For the model on activity density we used negative binomial distribution (log link function); for richness, evenness and Shannon entropy we used Gaussian distribution; and for inverse Simpson index we used gamma distribution (inverse link function). All models were tested for model fit using the DHARMA package.

Community composition of crop configurations and crops

To assess whether crop configurations have distinct ground beetle communities, we used permanova with a Hellinger transformation (with 999 permutations). We only used data from locations and years where pitfall catches had been identified to species level (Almere 2021/22, Lelystad and Wageningen). We considered four models, one for all locations combined and one for each location separately. In all models we included crop configuration and the nested variables of location (whenever applicable), year and crop as fixed factors. We also analysed the interaction between crop configuration and this nested structure of location, year and crop. To visualize crop-configuration-dependent ground beetle communities, we used redundancy analysis (RDA) on Hellinger transformed data. Here, we again conducted four analyses, one for all locations combined and one for each of the three considered locations. We only used crop configuration as a predictor, to force RDA to show any change in ground beetle community associated with crop configuration. As such, only one RDA axis was created per model, which was plotted against the first principal component describing the residual variation.

Due to the large influence of location and year on ground beetle communities, visualizing any general effects of strip cropping on these communities using all available data was challenging. To address this, we conducted RDA and visualized the effect of crop configuration on a subset of the data from one location and one year. We chose the data from Wageningen in 2021 and 2022 for this analysis because it provided a set-up where strip cropping of two crops could be compared with their constituent monocultures within the same experimental fields. Here, we used both crop configuration, crop and their interaction term as explanatory variables for each crop pair separately. To analyse whether ground beetle communities significantly differed among combinations of crops and crop configurations, we ran pairwise permanova on all three fields and two years separately, using the “pairwiseAdonis” package.

Crop configuration specific species

To assess whether there are crop-configuration-specific species we used indicator species analyses (ISA) (indicspecies package). Here, we excluded data from Almere in 2020, as they only contained genus-level identification. As this method does not allow for conditioning the data on location, we analysed the data for the different locations separately. To test whether species mainly occur in a monoculture or in strip cropping, we ran an ISA over the whole dataset per location. ISA uses two indicators: specificity and sensitivity of species to a certain crop configuration. Specificity refers to the predictive value of the species as indicator of the specific crop configuration, whereas sensitivity refers to the percentage of samples of the respective crop configuration that include the species. Both indicators do not take the abundance of species into account (de Cáceres, 2023 [↗](#)).

Statements

Funding

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Author contributions

Luuk Croijmans: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision. *Fogelina Cuperus*: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision. *Dirk F. van Apeldoorn*: Conceptualization, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. *Felix J.J.A. Bianchi*: Conceptualization, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition. *Walter A.H. Rossing*: Conceptualization, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition. *Erik H. Poelman*: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Ethics approval

This research was not reviewed by an institutional or governmental regulatory body as the work was performed on invertebrates.

Availability of data and materials

Data will be made publicly available upon acceptance in the 4TU repository.

Supplementary figures

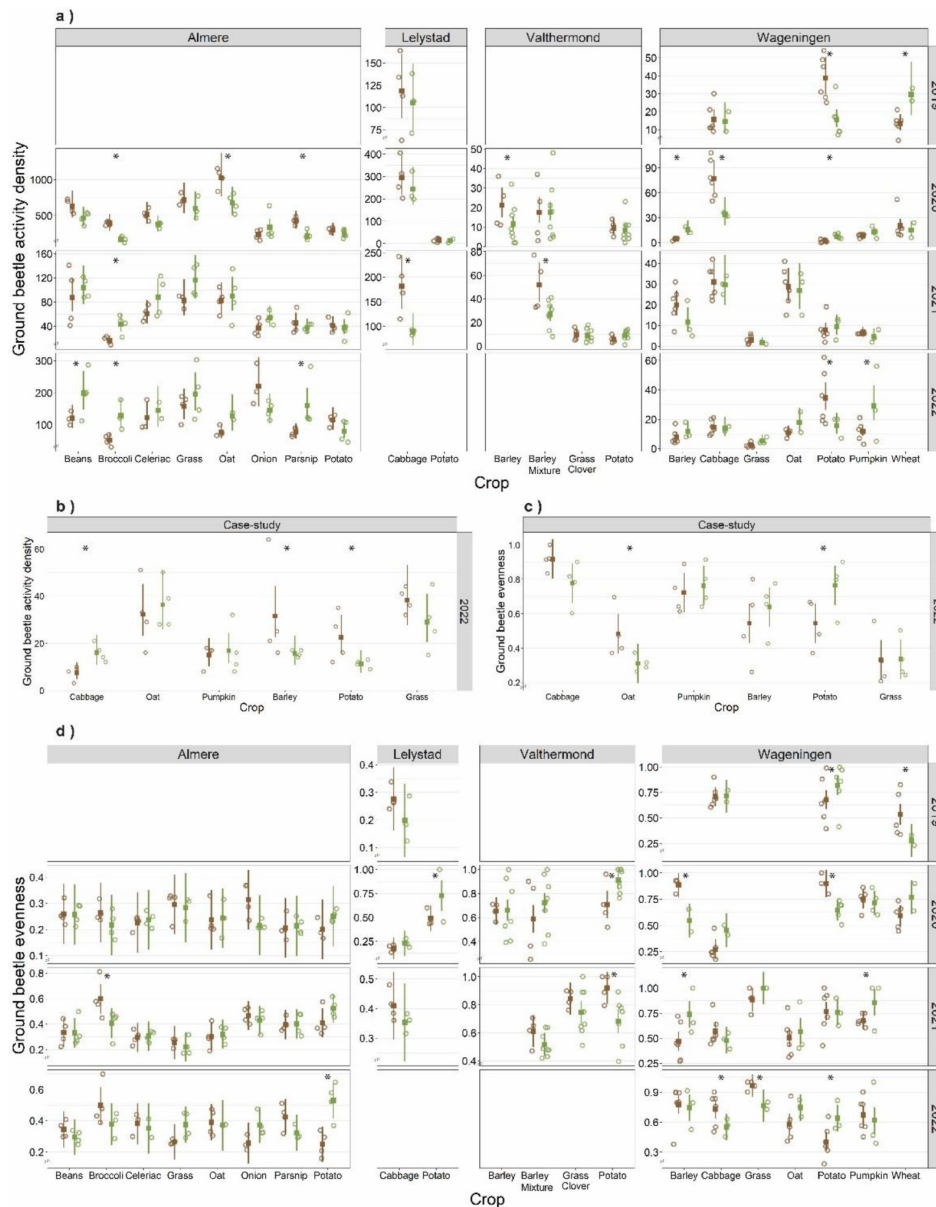


Fig. S1.

Effect of crop configuration on ground beetle activity density and absolute evenness.

Effect of crop configuration on activity density **(a)** and absolute evenness **(d)** per combination of location (columns), year (rows) and crop (x-axis); and the effect of crop configuration on activity density **(b)** and absolute evenness **(c)** in the case-study in Wageningen, consisting of two early spring sampling rounds in 2022. Activity density here is the total number of ground beetles captured in year series, which might differ in sampling frequency ([Table S1](#)). Absolute evenness is calculated from the total number of ground beetles captured in year series. Empty panels indicate combinations of years and locations that were not sampled. Squares indicate estimated means, the bar indicates the 95% confidence interval. When no estimated mean and confidence interval are shown, then those crops were not grown or sampled in that year. Asterisks indicate significant differences among the crop configurations ($\alpha = 0.05$). Open circles indicate individual year series ([Table S1](#)).

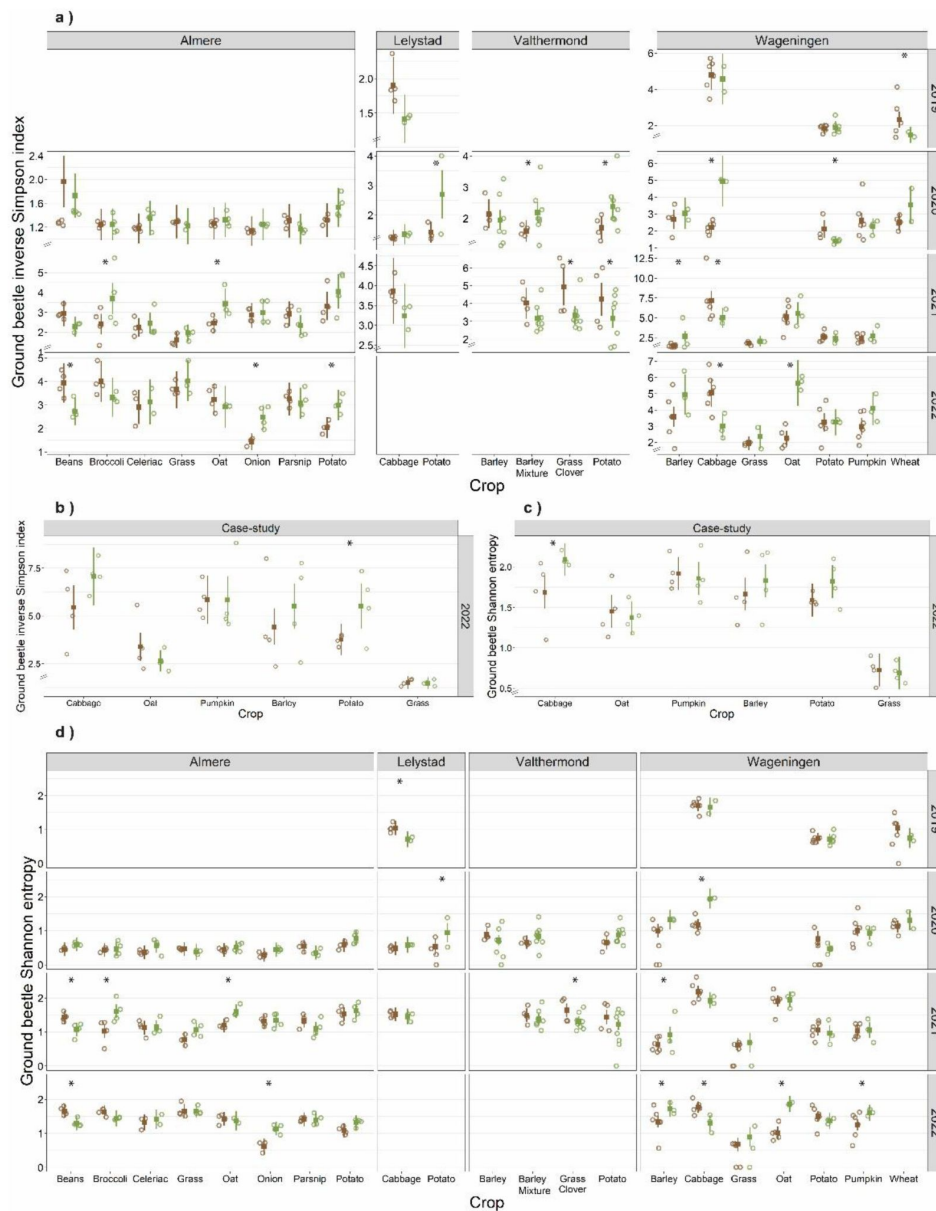


Fig. S2.

Effect of crop configuration on ground beetle inverse Simpson index and Shannon entropy.

Effect of crop configuration on inverse Simpson index (**a**) and Shannon entropy (**d**) per combination of location (columns), year (rows) and crop (x-axis); and the effect of crop configuration on inverse Simpson index (**b**) and Shannon entropy (**c**) in the case-study in Wageningen, consisting of two early spring sampling rounds in 2022. Inverse Simpson index and Shannon entropy were calculated from the total number of ground beetles captured in year series. Empty panels indicate combinations of years and locations that were not sampled. Squares indicate estimated means, the bar indicates the 95% confidence interval. When no estimated mean and confidence interval are shown, then those crops were not grown or sampled in that year. Asterisks indicate significant differences among the crop configurations ($\alpha = 0.05$). Open circles indicate individual year series ([Table S1](#)).

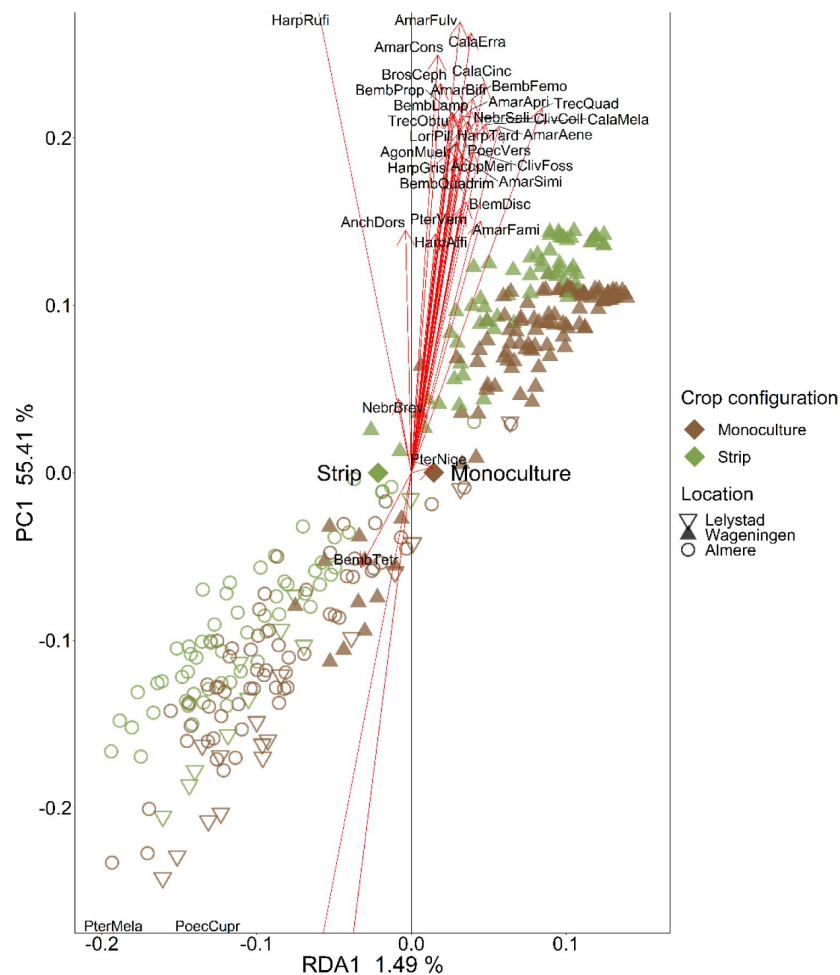


Fig. S3.

The effect of crop configuration on ground beetle community composition.

Data from all datasets that included species level data (Almere (○), Lelystad (▽), Wageningen (▲)) are used for visualizing the first RDA and PC axis. Each axis shows the percentage explained variation. Colour of the dots indicates crop configuration (brown = monoculture, green = strip cropping), and the direction of each crop configuration on the RDA axis (x-axis) is mentioned. Red arrows indicate species placement and name codes are given close to the tip of the arrow, meaning of the name codes can be found in [Table S3](#).

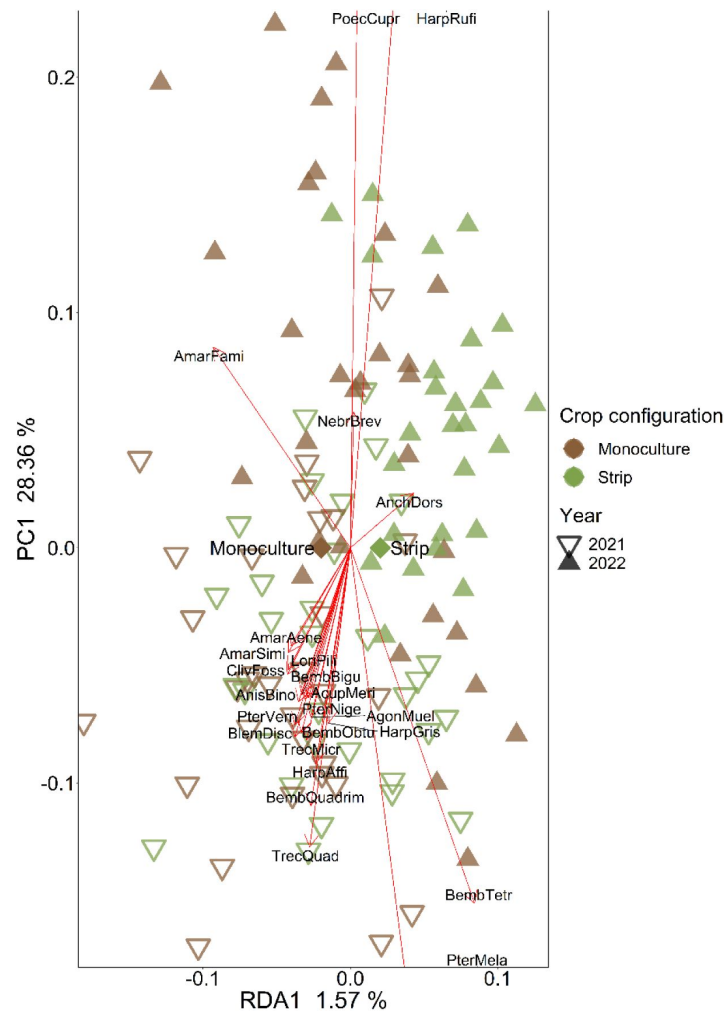


Fig. S4.

The effect of crop configuration on ground beetle community composition in Almere.

The first RDA and PC axis are visualized. Each axis shows the percentage explained variation. Shape of the dots indicates year (∇ = 2021, $\blacktriangle$ = 2022), colour of the dots indicates crop configuration (brown = monoculture, green = strip cropping), and the direction of each crop configuration on the RDA axis (x-axis) is mentioned. Red arrows indicate species placement and name codes are given close to the tip of the arrow, meaning of the name codes can be found in [Table S3](#).

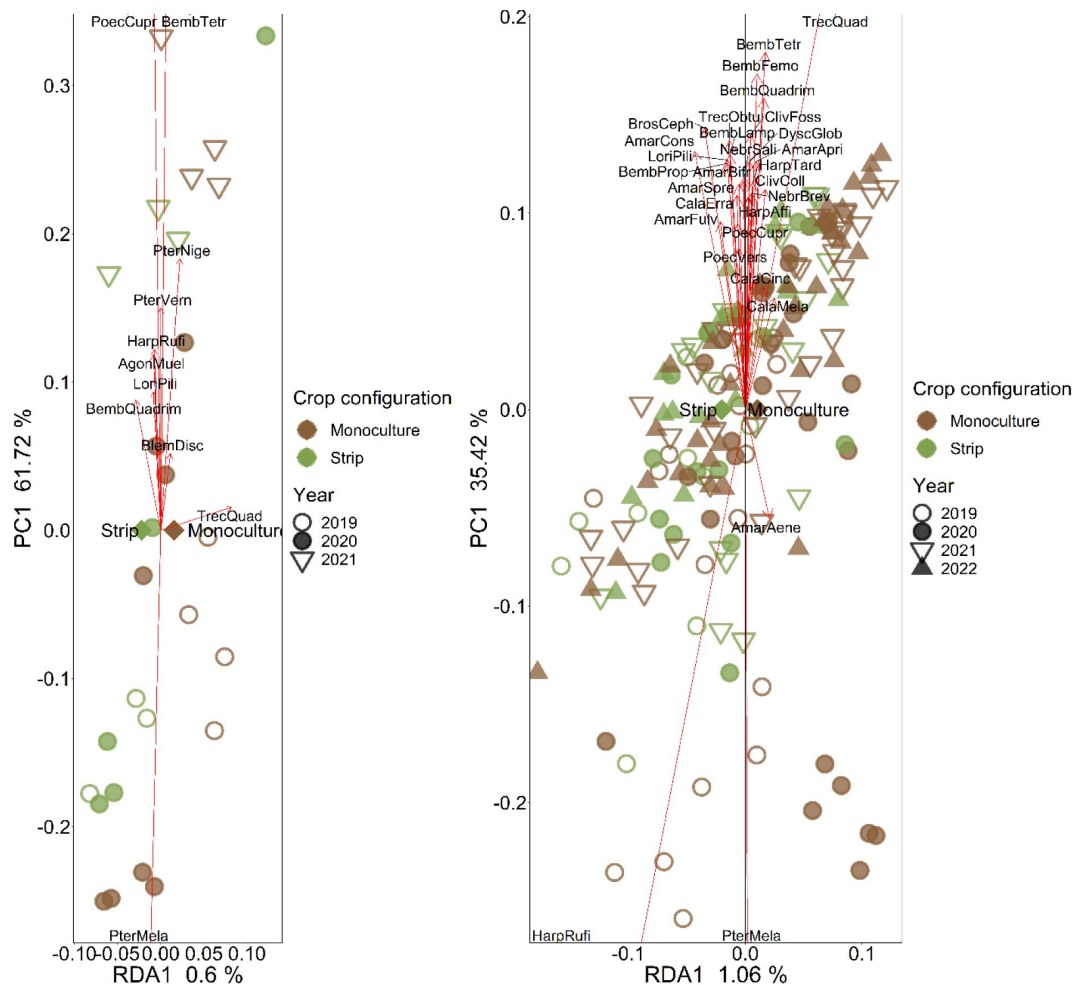


Fig. S5.

The effect of crop configuration on ground beetle community composition in Lelystad (left) and Wageningen (right).

The first RDA and PC axis are visualized. Each axis shows the percentage explained variation. Shape of the dots indicates year (○ = 2019, ● = 2020, ▽ = 2021, ▲ = 2022), colour of the dots indicates crop configuration (brown = monoculture, green = strip cropping), and the direction of each crop configuration on the RDA axis (x-axis) is mentioned. Red arrows indicate species placement and name codes are given close to the tip of the arrow, meaning of the name codes can be found in [Table S3](#).

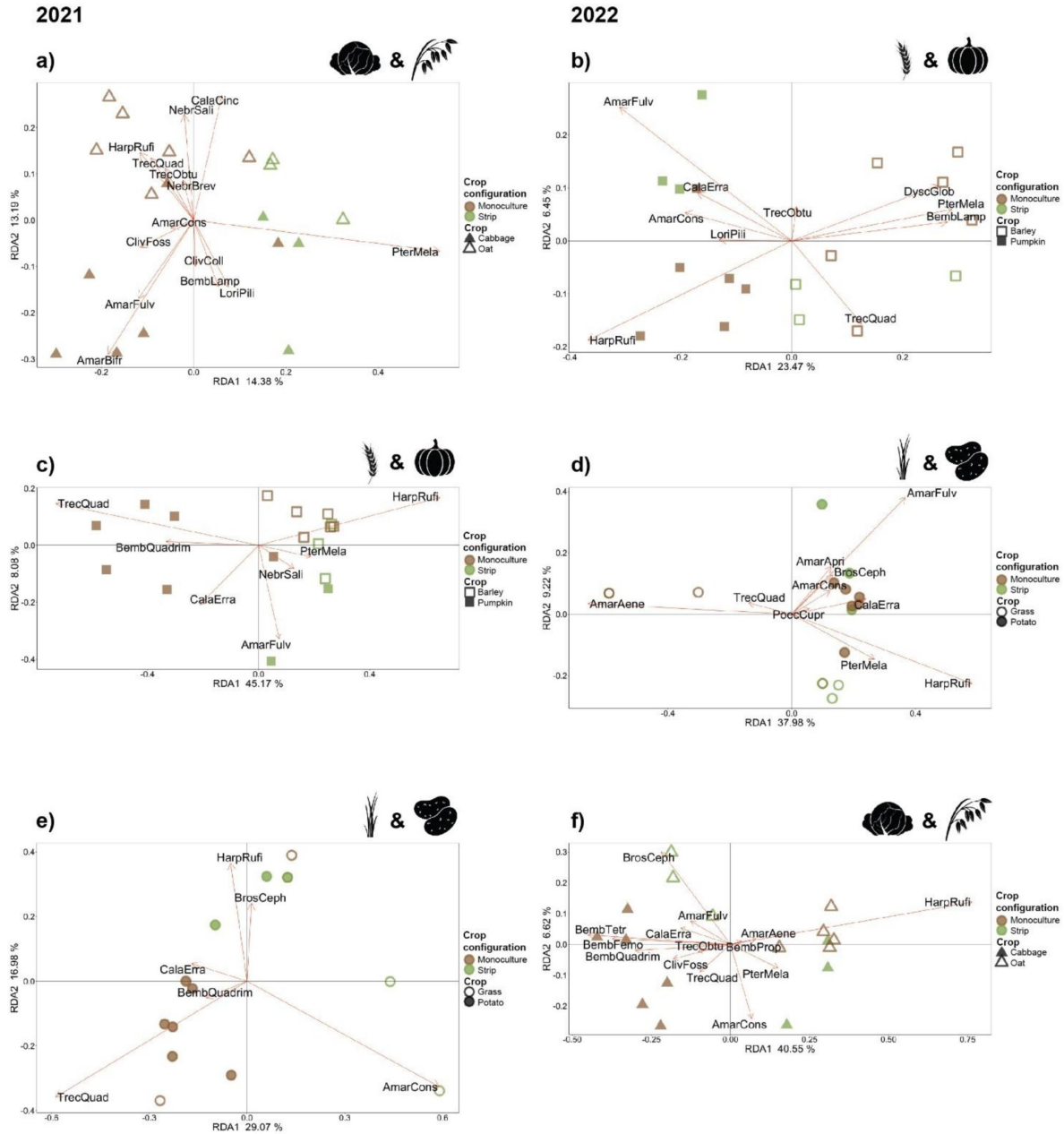


Fig. S6.

Effect of crop configuration and crop on ground beetle community composition in Wageningen.

Here, we use data from Wageningen in 2021 (left column; panels a, c, e) and 2022 (right column; panels b, d, f) from three fields including three crop pairs being cabbage and oat (panel a and f), barley and pumpkin (panel b and c), and grass and potato (panel d and e). The first and second RDA axis are visualized. Each axis shows the percentage explained variation. Colour of the dots indicates crop configuration (brown = monoculture, green = strip cropping), shapes indicates crop ($\square$ = barley, $\circ$ = pumpkin, $\bullet$ = potato, $\triangle$ = oat, $\blacktriangle$ = cabbage). The red lines and abbreviated names indicate how specific ground beetles species correlate with the RDA axes, species names are given in [Table S3](#).

A Almere

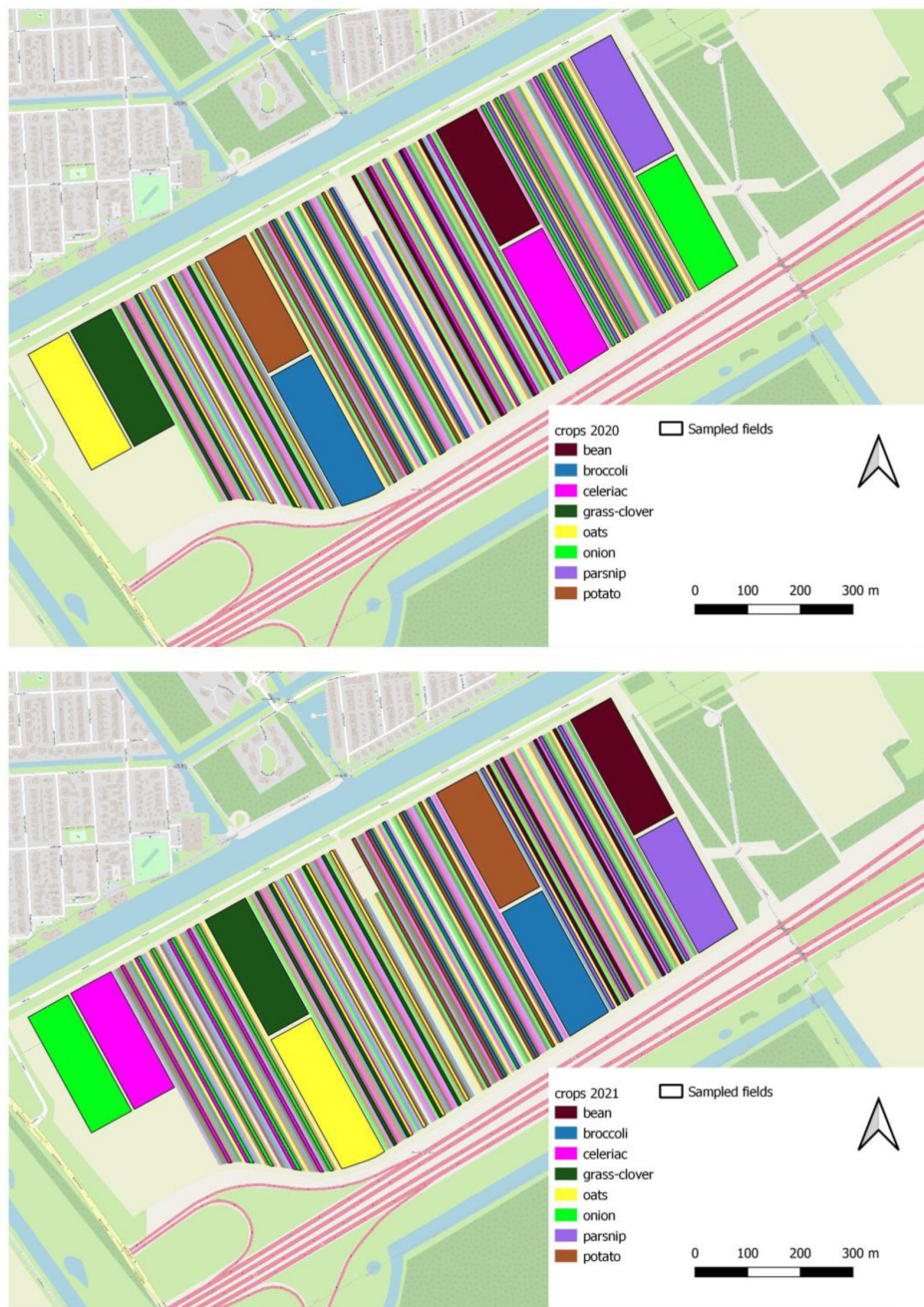
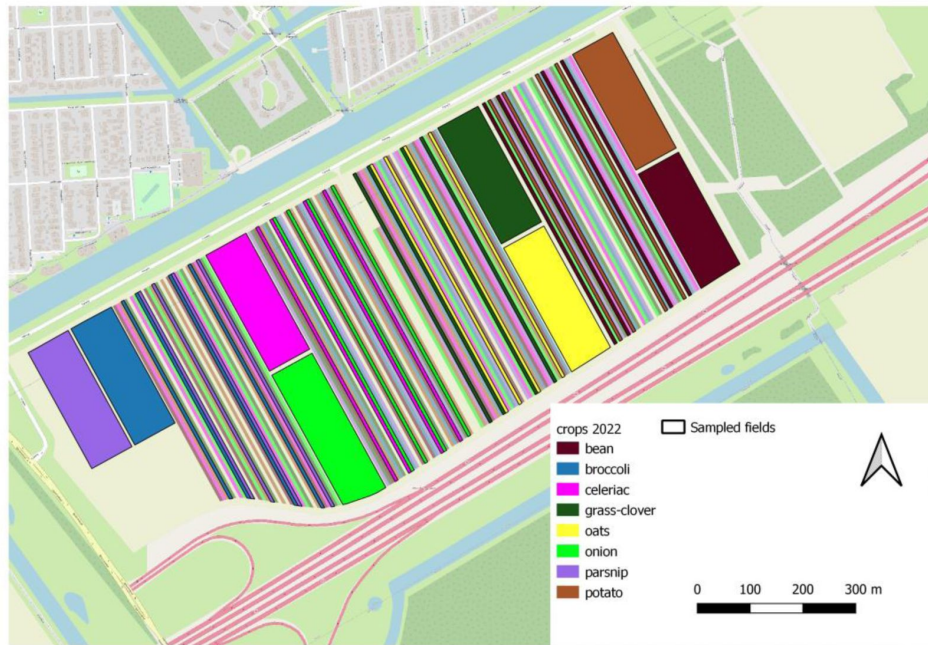


Fig. S7.

Field maps of experimental lay-out per year and location.

Field maps are shown in chronological order (oldest to newest) per location: Almere **(a)**, Lelystad **(b)**, Valthermond **(c)** and Wageningen **(d)**. Colors indicate distinct crops, whereas diagonal lines indicate a crop mixture (only for Valthermond). Brighter colours with black lining around the field indicate areas in which samples included in this study were taken.



B Lelystad

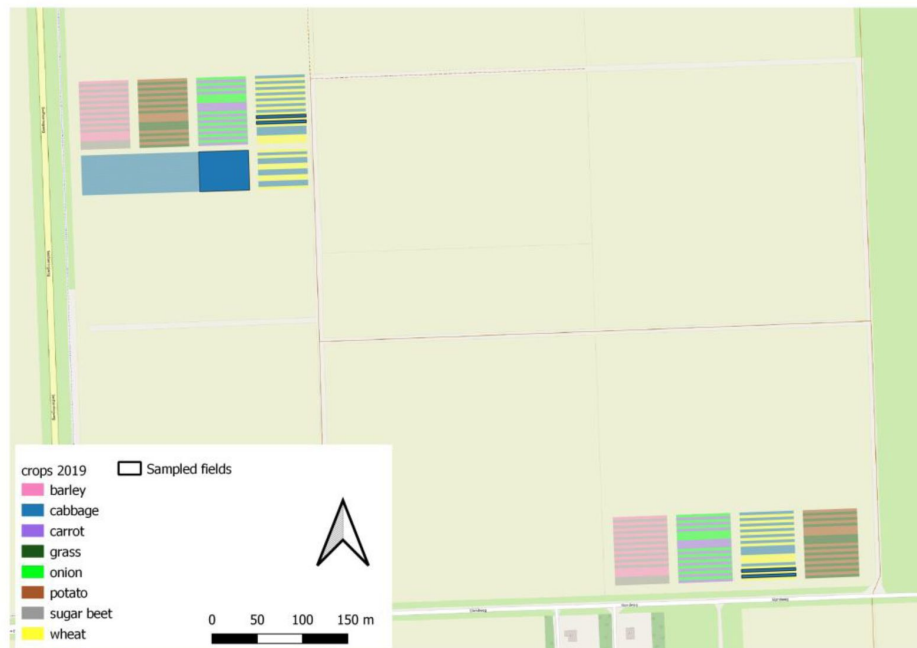


Fig. S7. (continued)



Fig. S7. (continued)

C Valthermond

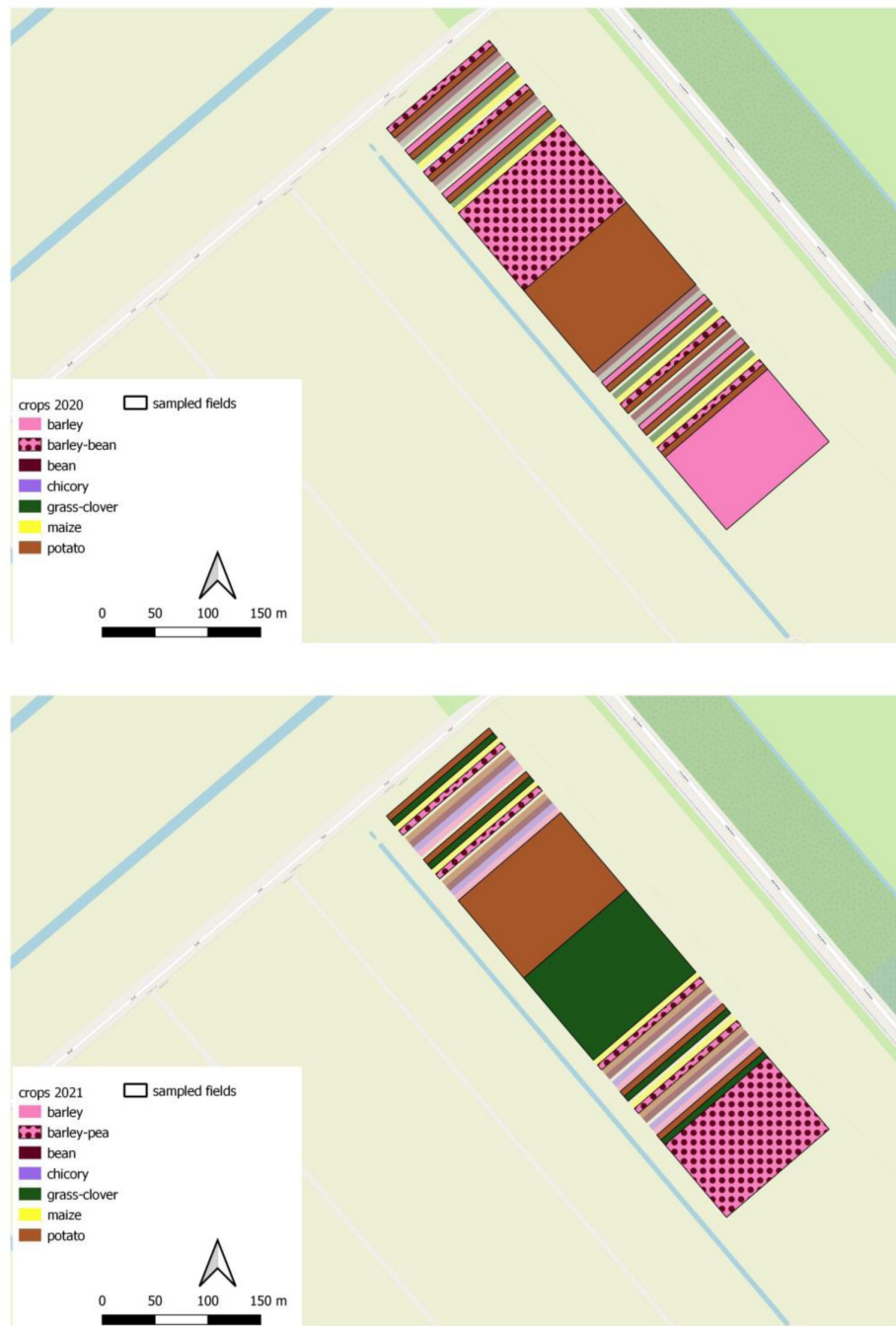


Fig. S7. (continued)

D Wageningen

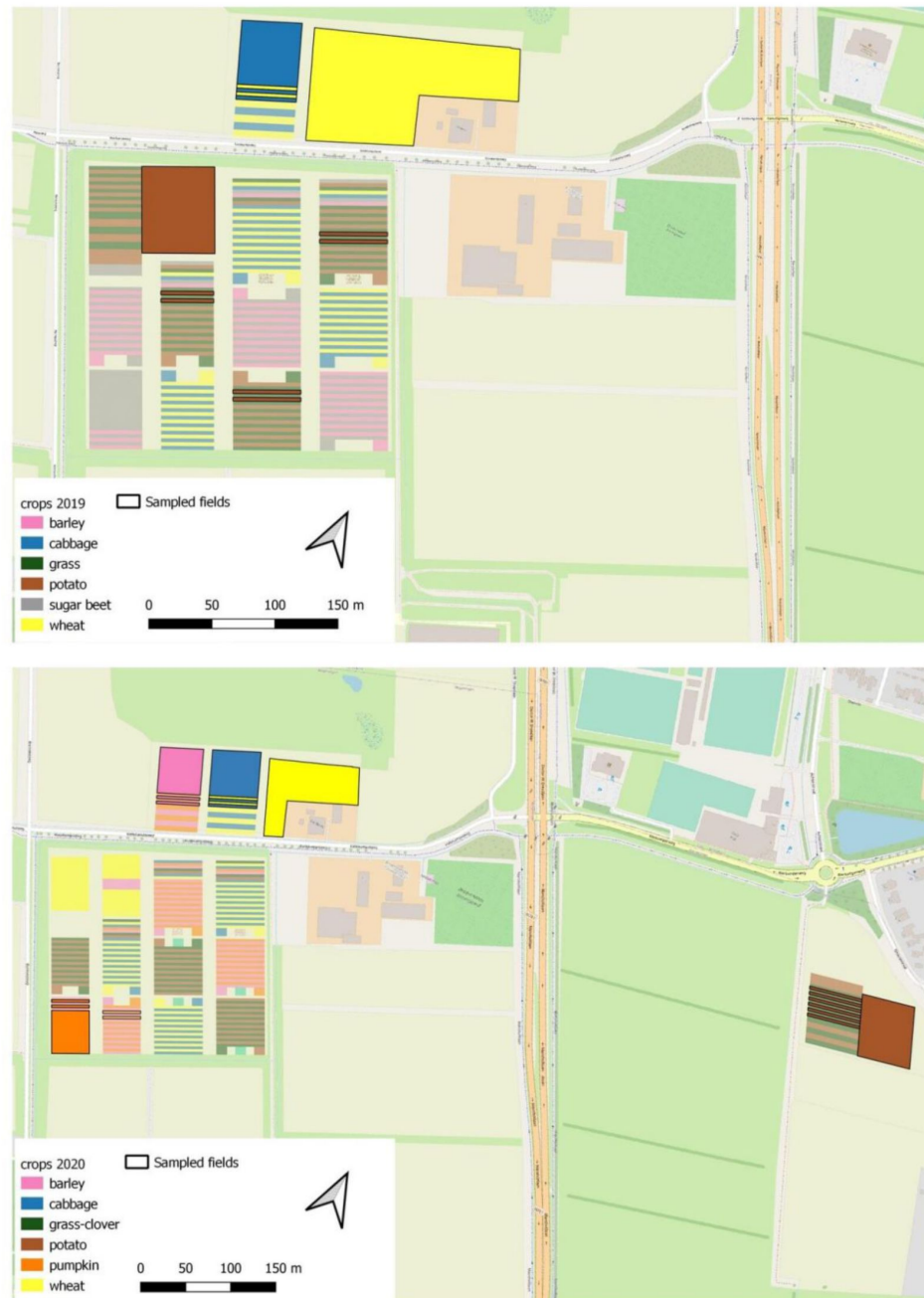


Fig. S7. (continued)



Fig. S7. (continued)

Supplementary tables

Acknowledgements

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Subfamily	Genus	Species	Species code
Carabinae	Carabus	<i>Carabus granulatus</i>	CaraGran
	Nebria	<i>Nebria brevicollis</i>	NebrBrev
		<i>Nebria salina</i>	NebrSali
	Notiophilus	<i>Notiophilus aquaticus</i>	NotiAqua
		<i>Notiophilus biguttatus</i>	NotiBigu
		<i>Notiophilus palustris</i>	NotiPalu
Harpalinae	Acupalpus	<i>Acupalpus meridianus</i>	AcupMeri
	Agonum	<i>Agonum muelleri</i>	AgonMuel
	Amara	<i>Amara aenea</i>	AmarAene
		<i>Amara anthobia</i>	AmarAnth
		<i>Amara apricaria</i>	AmarApri
		<i>Amara aulica</i>	AmarAuli
		<i>Amara bifrons</i>	AmarBifr
		<i>Amara communis</i>	AmarComm
		<i>Amara consularis</i>	AmarCons
		<i>Amara famelica</i>	AmarFame
		<i>Amara familiaris</i>	AmarFami
		<i>Amara fulva</i>	AmarFulv
		<i>Amara ovata</i>	AmarOvat
		<i>Amara plebeja</i>	AmarPleb
		<i>Amara similata</i>	AmarSimi
		<i>Amara spreta</i>	AmarSpre
		<i>Amara tibialis</i>	AmarTibi
	Anchomenus	<i>Anchomenus dorsalis</i>	AnchDors
	Anisodactylus	<i>Anisodactylus binotatus</i>	AnisBino
	Badister	<i>Badister bullatus</i>	BadiBull
		<i>Badister sodalist</i>	BadiSoda
	Bradycellus	<i>Bradycellus harpalinus</i>	BradHarp
	Calathus	<i>Calathus cinctus</i>	CalaCinc

Table S1.

All ground beetle species found among the four locations, and their species codes as used in several figures.

		<i>Calathus erratus</i>	CalaErra
		<i>Calathus fuscipes</i>	CalaFusc
		<i>Calathus melanocephalus</i>	CalaMela
		<i>Calathus rotundicollis</i>	CalaRotu
	Harpalus	<i>Harpalus affinis</i>	HarpAffi
		<i>Harpalus distinguendus</i>	HarpDist
		<i>Harpalus griseus</i>	HarpGris
		<i>Harpalus rubripes</i>	HarpRubr
		<i>Harpalus rufipes</i>	HarpRufi
		<i>Harpalus signaticornis</i>	HarpSign
		<i>Harpalus tardus</i>	HarpTard
	Microlestes	<i>Microlestes minutulus</i>	MicrMinu
	Oxypselaphus	<i>Oxypselaphus obscurus</i>	OxypObsc
	Panagaeus	<i>Panagaeus bipustulatus</i>	PanaBipu
	Poecilus	<i>Poecilus cupreus</i>	PoecCupr
		<i>Poecilus versicolor</i>	PoecVers
	Pterostichus	<i>Pterostichus anthracinus</i>	PterAnth
		<i>Pterostichus melanarius</i>	PterMela
		<i>Pterostichus niger</i>	PterNige
		<i>Pterostichus strenuus</i>	PterStre
		<i>Pterostichus vernalis</i>	PterVern
	Stenolophus	<i>Stenolophus teutonius</i>	StenTeut
	Stomis	<i>Stomis pumicatus</i>	StomPumi
	Syntomus	<i>Syntomus truncatellus</i>	SyntTrun
Loricarinae	Loricera	<i>Loricera pilicornis</i>	LoriPili
Scaritinae	Broschus	<i>Broschus cephalotes</i>	BrosCeph
	Clivina	<i>Clivina collaris</i>	ClivColl
		<i>Clivina fossor</i>	ClivFoss
	Dyschirius	<i>Dyschirius globosus</i>	DyscGlob
Trechinae	Asaphidion	<i>Asaphidion flavipes</i>	AsapFlav
	Bembidion	<i>Bembidion aenea</i>	BembAene
		<i>Bembidion biguttatum</i>	BembBigu
		<i>Bembidion femoratum</i>	BembFemo

Table S1. (continued)

	<i>Bembidion lampros</i>	BembLamp
	<i>Bembidion lunulatum</i>	BembLunu
	<i>Bembidion obtusum</i>	BembObtu
	<i>Bembidion proprans</i>	BembProp
	<i>Bembidion quadrimaculatum</i>	BembQuadrim
	<i>Bembidion tetracolum</i>	BembTetr
Blemus	<i>Blemus discus</i>	BlemDisc
Trechoblemus	<i>Trechoblemus micros</i>	TrecMicr
Trechus	<i>Trechus obtusus</i>	TrecObtu
	<i>Trechus quadristriatus</i>	TrecQuad

Table S1. (continued)

	Species	Al	Le	Va	Wa	Total
1	<u><i>Pterostichus sp.</i></u>	25258	N/A	68	N/A	25326
2	<i>Pterostichus melanarius</i>	5556	2731	144	765	9196
3	<i>Poecilus cupreus</i>	2876	233	N/A	66	3175
4	<i>Harpalus rufipes</i>	1409	15	168	1143	2735
5	<u><i>Poecilus sp.</i></u>	1769	N/A	82	N/A	1851
6	<u><i>Harpalus sp.</i></u>	947	N/A	308	N/A	1255
7	<i>Bembidion tetracolum</i>	345	397	N/A	44	786
8	<i>Amara aenea</i>	15	0	8	654	677
9	<i>Pterostichus niger</i>	167	228	5	1	401
10	<i>Nebria brevicollis</i>	320	2	0	24	346
11	<u><i>Bembidion sp.</i></u>	300	N/A	32	N/A	332
12	<i>Anchomenus dorsalis</i>	151	1	161	8	321
13	<u><i>Trechus sp.</i></u>	303	N/A	N/A	N/A	303
14	<i>Trechus quadristriatus</i>	69	76	0	137	282
15	<i>Amara familiaris</i>	124	0	0	97	221
16	<i>Harpalus affinis</i>	131	3	1	57	192

Table S2.

Total number of ground beetles caught per species (or genus), per location. For some locations, ground beetles were identified up to genus level, these are underlined. "N/A" indicates that this taxa was identified to a different taxonomic level for the specific location. Locations are indicated with abbreviations (Al=Almere; Le = Lelystad; Va = Valthermond; Wa = Wageningen).

17	<i>Bembidion quadrimaculatum</i>	87	8	N/A	68	163
18	<i>Clivina fossor</i>	49	9	2	66	126
19	<i>Calathus melanocephalus</i>	0	0	67	42	109
20	<i>Amara fulva</i>	0	0	6	98	104
21	<i>Calathus cinctus</i>	0	0	20	73	93
22	<i>Calathus erratus</i>	0	0	6	87	93
23	<i>Poecilus versicolor</i>	4	2	N/A	80	86
24	<i>Loricera pilicornis</i>	22	6	20	37	85
25	<i>Blemus discus</i>	34	45	0	0	79
26	<i>Pterostichus vernalis</i>	57	9	N/A	6	72
27	<i>Amara spreta</i>	0	0	0	71	71
28	<i>Bembidion proprans</i>	4	0	N/A	64	68
29	<i>Clivina collaris</i>	0	2	2	64	68
30	<i>Amara consularis</i>	0	0	0	54	54
31	<i>Bembidion lampros</i>	3	1	N/A	44	48
32	<i>Nebria sp.</i>	42	N/A	N/A	N/A	42
33	<i>Amara apricaria</i>	0	0	0	41	41
34	<i>Amara bifrons</i>	1	0	1	36	38
35	<i>Bembidion femoratum</i>	0	0	N/A	36	36
36	<i>Broscus cephalotes</i>	0	0	0	35	35
37	<i>Amara similata</i>	24	0	0	6	30
38	<i>Harpalus tardus</i>	0	0	2	26	28
39	<i>Agonum muelleri</i>	5	3	7	11	26
40	<i>Nebria salina</i>	0	0	0	23	23
41	<i>Trechus obtusus</i>	2	0	0	18	20
42	<i>Harpalus griseus</i>	9	0	2	4	15
43	<i>Acupalpus meridianus</i>	8	3	0	3	14
44	Unknown carabidae	10	1	0	3	14
45	<i>Amara ovata</i>	0	0	9	1	10
46	<i>Dyschirius globosus</i>	0	0	0	10	10
47	<i>Anisodactylus binotatus</i>	6	0	0	2	8
48	<i>Amara communis</i>	0	0	0	7	7
49	<i>Bembidion biguttatum</i>	7	0	N/A	0	7
50	<i>Amara sp.</i>	5	N/A	N/A	N/A	5

Table S2. (continued)

51 <i>Clivina sp.</i>	5	N/A	N/A	N/A	5
52 <i>Harpalus distinguendus</i>	0	0	N/A	5	5
53 <i>Harpalus signaticornis</i>	0	0	N/A	5	5
54 <i>Bembidion obtusum</i>	4	0	N/A	0	4
55 <i>Trechoblemus micros</i>	4	0	0	0	4
56 <i>Amara anthobia</i>	0	0	3	0	3
57 <i>Badister bullatus</i>	2	0	0	1	3
58 <i>Badister sodalis</i>	3	0	0	0	3
59 <i>Bembidion lunulatum</i>	3	0	N/A	0	3
60 <i>Oxytelus obscurus</i>	3	0	0	0	3
61 <i>Stomis pumicatus</i>	3	0	0	0	3
62 <i>Acupalpus sp.</i>	2	N/A	N/A	N/A	2
63 <i>Amara aulica</i>	1	0	0	1	2
64 <i>Calathus fuscipes</i>	0	0	2	0	2
65 <i>Calathus rotundicollis</i>	0	0	0	2	2
66 <i>Carabus granulatus</i>	2	0	0	0	2
67 <i>Harpalus rubripes</i>	0	1	N/A	1	2
68 <i>Notiophilus palustris</i>	2	0	0	0	2
69 <i>Pterostichus strenuus</i>	2	0	N/A	0	2
70 <i>Stenolophus teutonius</i>	0	0	0	2	2
71 <i>Amara famelica</i>	0	0	0	1	1
72 <i>Amara plebeja</i>	1	0	0	0	1
73 <i>Amara tibialis</i>	0	0	0	1	1
74 <i>Asaphidion flavipes</i>	0	0	0	1	1
75 <i>Bembidion aenea</i>	0	1	N/A	0	1
76 <i>Bradycellus harpalinus</i>	1	N/A	N/A	N/A	1
77 <i>Carabus sp.</i>	1	N/A	N/A	N/A	1
78 <i>Microlestes minutulus</i>	0	0	0	1	1
79 <i>Notiophilus aquaticus</i>	0	0	0	1	1
80 <i>Notiophilus biguttatus</i>	0	0	0	1	1
81 <i>Panagaeus bipustulatus</i>	0	0	0	1	1
82 <i>Pterostichus anthracinus</i>	1	0	N/A	0	1
83 <i>Syntomus truncatellus</i>	0	0	0	1	1

Table S2. (continued)

Location	Predictor	Df	Sum Sq	R2	F	P
Almere	Crop configuration	1	0.20	0.01	2.34	0.025
	Year	1	2.48	0.13	28.6	0.001
	Year : Crop species	14	5.66	0.30	4.68	0.001
	Crop configuration : Year	1	0.13	0.01	1.50	0.138
	Crop configuration : Year : Crop species	14	2.88	0.15	2.38	0.001
	<i>Residual</i>	88	<i>7.61</i>	<i>0.40</i>		
	<i>Total</i>	119	<i>19.0</i>	<i>1.00</i>		
Lelystad	Crop configuration	1	0.03	0.01	0.68	0.534
	Year	2	1.78	0.55	18.9	0.001
	Year : Crop species	1	0.12	0.04	2.45	0.082
	Crop configuration : Year	2	0.26	0.08	2.80	0.026
	Crop configuration : Year : Crop species	1	0.14	0.04	2.91	0.039
	<i>Residual</i>	19	<i>0.90</i>	<i>0.28</i>		
	<i>Total</i>	26	<i>3.22</i>	<i>1.00</i>		
Wageningen	Crop configuration	1	1.48	0.02	4.30	0.001
	Year	3	6.40	0.07	6.22	0.001
	Year : Crop species	16	27.7	0.29	5.05	0.001
	Crop configuration : Year	3	1.90	0.02	1.85	0.006
	Crop configuration : Year : Crop species	16	11.8	0.12	2.15	0.001
	<i>Residual</i>	138	<i>47.3</i>	<i>0.49</i>		
	<i>Total</i>	177	<i>96.6</i>	<i>1.00</i>		

Table S3.

Effect of crop configuration on ground beetle community composition.

Results from permanova analyses using Hellinger's transformation for data from the three locations with species level data. "Crop species" is a nested variable within years, as these differed among years. Years were nested in locations, as the years that were studied differed among locations. P-values in bold typeset indicate significant effects ($\alpha = 0.05$).

Table S4a. 2021

F-value		Pumpkin		Barley	
		Mono	Strip	Mono	Strip
Pumpkin	Mono				
	Strip	3.64			
Barley	Mono	10.39	1.40		
	Strip	5.82	1.21	4.23	

F-value		Potato		Grass	
		Mono	Strip	Mono	Strip
Potato	Mono				
	Strip	5.00			
Grass	Mono	1.70	0.57		
	Strip	8.38	5.06	1.94	

F-value		Cabbage		Oat	
		Mono	Strip	Mono	Strip
Cabbage	Mono				
	Strip	1.47			
Oat	Mono	2.08	3.52		
	Strip	2.06	2.07	2.92	

Table S4.

Effect of crop configuration and crop species on ground beetle community composition.

Results from pairwise permanova analyses for crop pairs pumpkin-barley, cabbage-oat, and potato-grass in 2021 (a) and 2022 (b) in Wageningen. Values show F-values for the comparison between the crop configurations and crops in crossing rows and columns. Bold numbers indicate significant differences between combinations of crop configurations and crops ($\alpha = 0.05$).

Table S4b. 2022

F-value		Pumpkin		Barley	
		Mono	Strip	Mono	Strip
Pumpkin	Mono				
	Strip	1.42			
Barley	Mono	3.84	2.90		
	Strip	1.82	2.18	0.76	

F-value		Potato		Grass	
		Mono	Strip	Mono	Strip
Potato	Mono				
	Strip	1.42			
Grass	Mono	7.14	3.26		
	Strip	4.01	3.24	3.58	

F-value		Cabbage		Oat	
		Mono	Strip	Mono	Strip
Cabbage	Mono				
	Strip	5.35			
Oat	Mono	8.72	1.82		
	Strip	1.59	3.56	5.05	

Table S4. (continued)

Almere					
Species	Specificity	Sensitivity	Stat	p	
MONO					
<i>Amara aenea</i>	0.733	0.219	0.401	0.031	*
<i>Blemus discus</i>	0.833	0.156	0.361	0.030	*
<i>Anisodactylus binotatus</i>	0.833	0.094	0.280	0.182	
<i>Badister sodalis</i>	0.750	0.094	0.265	0.281	
<i>Bembidion obtusum</i>	0.750	0.094	0.265	0.274	
STRIP					
<i>Anchomenus dorsalis</i>	0.806	0.656	0.727	0.001	***
<i>Harpalus griseus</i>	0.900	0.250	0.474	0.001	***
<i>Bembidion lunulatum</i>	1	0.094	0.306	0.040	*
<i>Amara aulica</i>	1	0.031	0.177	0.496	
<i>Amara bifrons</i>	1	0.031	0.177	0.511	
<i>Amara plebeja</i>	1	0.031	0.177	0.498	
<i>Pterostichus anthracinus</i>	1	0.031	0.177	0.515	
Lelystad					
Species	Specificity	Sensitivity	Stat	p	
MONO					
<i>Trechus quadristriatus</i>	0.774	0.325	0.502	0.006	*
<i>Pterostichus vernalis</i>	0.798	0.063	0.223	0.195	
<i>Poecilus versicolor</i>	1	0.025	0.158	0.222	
<i>Bembidion aeneum</i>	1	0.013	0.112	0.485	
STRIP					
<i>Anchomenus dorsalis</i>	1	0.044	0.211	0.128	

Table S5.

Effect of crop configuration on ground beetle indicator species.

Results from indicator species analyses with the four locations being analysed separately. Data from Almere in 2020 data was excluded from the analysis as ground beetles were identified up to genus level in 2020. Specificity refers to the predictive value of the species as indicator of the specific crop configuration whereas sensitivity refers to the percentage of samples of the respective crop configuration that included the species. "Stat" refers to the indicator value, which is the square root of the product of specificity and sensitivity. Bold letters indicate a significant association between species and crop configurations per location ($\alpha = 0.05$).

<i>Harpalus affinis</i>	1	0.033	0.183	0.247
<i>Nebria brevicollis</i>	1	0.022	0.149	0.539
<i>Amara aenea</i>	1	0.011	0.105	1
<i>Amara familiaris</i>	1	0.011	0.105	1
<i>Bembidion lampros</i>	1	0.011	0.105	1
<i>Harpalus rubripes</i>	1	0.011	0.105	1
<i>Badister sodalis</i>	1	0.011	0.105	1

Valthermond

Species	Specificity	Sensitivity	Stat	p	
MONO					
<i>Loricera pilicornis</i>	0.750	0.292	0.468	0.044	*
<i>Amara aenae</i>	0.769	0.167	0.358	0.138	
<i>Clivina fossor</i>	1.000	0.083	0.289	0.103	
<i>Amara anthobia</i>	0.800	0.083	0.258	0.278	
<i>Amara bifrons</i>	1.000	0.042	0.204	0.341	
<i>Harpalus affinis</i>	1.000	0.042	0.204	0.354	
<i>Clivina collaris</i>	0.667	0.042	0.167	1.000	
<i>Harpalus tardus</i>	0.667	0.042	0.167	1.000	

STRIP

<i>Amara ovata</i>	1.000	0.188	0.433	0.042	*
<i>Harpalus griseus</i>	1.000	0.042	0.204	0.548	
<i>Calathus fuscipes</i>	1.000	0.042	0.204	0.553	
<i>Amara fulva</i>	1.000	0.021	0.144	1.000	

Wageningen

Species	Specificity	Sensitivity	Stat	p	
MONO					
<i>Calathus melanocephalus</i>	0.823	0.063	0.228	0.004	**
<i>Amara apricaria</i>	0.792	0.036	0.168	0.025	*
<i>Bembidion tetracolum</i>	0.782	0.036	0.167	0.083	.
<i>Harpalus tardus</i>	0.781	0.036	0.167	0.041	*
<i>Bembidion femoratum</i>	0.757	0.025	0.137	0.137	

Table S5. (continued)

<i>Harpalus signaticornis</i>	0.945	0.016	0.125	0.072	.
<i>Pterostichus vernalis</i>	1.000	0.014	0.117	0.056	.
<i>Anchomelis dorsalis</i>	1.000	0.014	0.117	0.068	.
<i>Dyschirius globosus</i>	0.792	0.016	0.114	0.195	
<i>Harpalus griseus</i>	0.740	0.008	0.078	0.621	
<i>Calathus rotundicollis</i>	1.000	0.005	0.074	0.507	
<i>Stenolophus teutonus</i>	1.000	0.005	0.074	0.508	
<i>Amara communis</i>	1.000	0.003	0.052	1.000	
<i>Acupalpus meridianus</i>	1.000	0.003	0.052	1.000	
<i>Badister bullates</i>	1.000	0.003	0.052	1.000	
<i>STRIP</i>					
<i>Harpalus distinguendus</i>	0.808	0.012	0.097	0.215	
<i>Pterostichus niger</i>	1.000	0.003	0.054	0.504	
<i>Amara tibialis</i>	1.000	0.003	0.054	0.488	
<i>Amara aulica</i>	1.000	0.003	0.054	0.488	
<i>Amara famelica</i>	1.000	0.003	0.054	0.512	
<i>Notiophilus aquaticus</i>	1.000	0.003	0.054	0.482	
<i>Demetrias atricapillus</i>	1.000	0.003	0.054	0.481	
<i>Bradycellus harpalinus</i>	1.000	0.003	0.054	0.492	
<i>Microlestes minutulus</i>	1.000	0.003	0.054	0.514	
<i>Badister sodalis</i>	1.000	0.003	0.054	0.483	

Table S5. (continued)

Location	Year	Crops	Reference	Yield (t/ha)	
				Mono	Strip
Almere	2020	Beans	Unpublished data from Juventia and van Apeldoorn, in preparation (see: attached manuscript under embargo)	2.80	3.45
		Broccoli		NC	NC
		Celeriac		54.44	69.57
		Grass		NC	NC
		Oat		6.05	5.08
		Onion		41.06	42.14
		Parsnip		27.50	35.36
		Potato		NC	NC
	2021	Beans	Unpublished data from Juventia and van Apeldoorn, in preparation (see: attached manuscript under embargo)	4.72	6.50
		Broccoli		9.30	4.90
		Celeriac		NC	NC
		Grass		NC	NC
		Oat		8.76	8.25
		Onion		39.03	37.39
		Parsnip		36.88	65.01
		Potato		24.68	27.88
	2022	Beans	Unpublished data from Juventia and van Apeldoorn, in preparation (see: attached manuscript under embargo)	4.12	4.36
		Broccoli		1.12	1.59
		Celeriac		45.38	51.45
		Grass		NC	NC
		Oat		6.27	6.70
		Onion		57.89	48.62
		Parsnip		20.6	21.09
		Potato		31.08	31.60
Lelystad	2020	Cabbage	Carillo-Reche et al., 2023	44.0 ± 12.8	46.3 ± 13.0
	2021	Cabbage	Carillo-Reche et al., 2023	85.6 ± 18.2	75.9 ± 23.4
Valthermond	2020	Barley	Unpublished data	5.3	5.1
		Barley / Beans		NC	5.1
		Potato		20.9	20.5
	2021	Barley / Beans	Unpublished data	3.1	3.4
		Grass / Clover		8.3	7.9

Table S6.

Effect of crop configuration on crop yield.

Yield results were retrieved from published and unpublished studies on effects of strip cropping on crop yield in similar locations, years, and crops as this study. Mean crop yield is presented in ton per hectare (t/ha). When known, standard deviations of mean crop yield are given ($\pm$ SD). When a crop is indicated with NC (not collected) the crop yield was not collected due to an inconsistent sampling method (potato, 2020, Almere), crop failure (broccoli, 2020, Almere; celeriac, 2021, Almere), unavailable machine-harvest data (grass, Almere, 2020, 2021, 2022) and undocumented reasons (barley/beans, 2020, Valthermond). Unavailable data include cabbage (2019) and potato (2020) in Lelystad; and pumpkin (2020, 2021, 2022), barley (2020, 2021, 2022), oat (2021, 2022), potato (2021, 2022), grass (2021, 2022), and cabbage (2022) in Wageningen.

		Potato		23.7	20.7
	2019	Cabbage	Carillo-Reche et al., 2023	36.8 ± 15.8	34.6 ± 6.5
		Wheat	Ditzler et al., 2023	2.8 ± 0.2	2.1 ± 0.3
		Potato	Ditzler et al., 2023	29.3 ± 5.7	30.6 ± 8.8
Wageningen	2020	Cabbage	Carillo-Reche et al., 2023	25.7 ± 9.3	19.3 ± 6.6
		Wheat	Ditzler et al., 2023	0.9 ± 0.2	1.0 ± 0.4
		Potato	Ditzler et al., 2023	23.4 ± 6.1	26.1 ± 8.9
	2021	Cabbage	Carillo-Reche et al., 2023	32.1 ± 7.1	27.9 ± 13.1

Table S6. (continued)

Location	Year	Crops	Rounds	Pitfall traps per round	
				Mono	Strip
Almere	2020	Beans	3	4	4
		Broccoli		4	4
		Celeriac		4	4
		Grass		4	3
		Oat		4	4
		Onion		4	4
		Parsnip		4	4
		Potato		4	4
	2021	Beans	4	4	4
		Broccoli		4	4
		Celeriac		4	4
		Grass		3	4
		Oat		4	4
		Onion		4	4
		Parsnip		4	4
		Potato		4	4
	2022	Beans	5	4	4
		Broccoli		4	3
		Celeriac		3	2
		Grass		4	4
		Oat		4	2
		Onion		3	4
		Parsnip		4	4
		Potato		4	4
Lelystad	2019	Cabbage	5	4	3
	2020	Cabbage	7	4	3

Table S7.

Sampling effort.

The total numbers of pitfall traps placed per location, year and crop. Rounds indicate the number of times pitfall traps were placed and were pooled within year series.

		Potato	2	4	2
	2021	Cabbage	6	4	3
Valthermond	2020	Barley	7	6	6
		Barley / Beans		6	6
		Potato		6	6
	2021	Barley / Beans	8	6	6
		Grass / Clover		6	6
		Potato		6	6
	Wageningen	2019	Cabbage	4	6
Wheat			2	6	2
Potato			3	6	6
2020		Cabbage	8	6	2
		Wheat	3	5	2
		Potato	1	6	5
		Pumpkin	2	6	4
		Barley		6	2
2021		Cabbage	7	6	3
		Oat		6	3
		Potato	2	6	3
		Grass		6	3
		Pumpkin		6	3
		Barley		6	3
2022		Cabbage	2	6	3
		Oat		5	3
		Potato		6	3
		Grass		6	3
	Pumpkin		6	3	
	Barley		6	3	

Table S7. (continued)

Species	
1	<i>Achillea millefolium</i> (common yarrow)
2	<i>Anethum graveolens</i> (dill)
3	<i>Cichorium intybus</i> (common chichory)
4	<i>Foeniculum vulgare</i> (fennel)
5	<i>Pastinaca sativa</i> (parsnip)
6	<i>Angelica sylvestris</i> (wild angelica)
7	<i>Anthriscus sylvestris</i> (cow parsley)
8	<i>Fagopyrum esculentum</i> (buckwheat)
9	<i>Papaver Rhoeas</i> (common poppy red wild form)
10	<i>Chrysanthemum segetum</i> (corn marigold)
11	<i>Ammi majus</i> (greater ammi)
12	<i>Centaurea cyanus</i> (cornflower, single flowered wild form)

Table S8.

Plant species composition of flower strips adjacent to the strip cropping fields in Lelystad and Valthermond (see location in Fig. S7).

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Reviewer #1 (Public review):

Summary:

This study demonstrates that strip cropping enhances the taxonomic diversity of ground beetles across organically-managed crop systems in the Netherlands. In particular, strip cropping supported 15% more ground beetle species and 30% more individuals compared to monocultures.

Strengths:

A well-written study with well-analyzed data of a complex design. The data could have been analyzed differently e.g. by not pooling samples, but there are pros and cons for each type of analysis and I am convinced this will not affect the main findings. A strong point is that data were collected for 4 years. This is especially strong as most data on biodiversity in cropping systems are only collected for one or two seasons. Another strong point is that several crops were included.

Weaknesses:

This study focused on the biodiversity of ground beetles and did not examine crop productivity. Therefore, I disagree with the claim that this study demonstrates biodiversity enhancement without compromising yield. The authors should present results on yield or, at the very least, provide a stronger justification for this statement.

<https://doi.org/10.7554/eLife.104762.1.sa2>

Reviewer #2 (Public review):

Summary:

The authors aimed to investigate the effects of organic strip cropping on carabid richness and density as well as on crop yields. They find on average higher carabid richness and density in strip cropping and organic farming, but not in all cases.

Strengths:

Based on highly resolved species-level carabid data, the authors present estimates for many different crop types, some of them rarely studied, at the same time. The authors did a great job investigating different aspects of the assemblages (although some questions remain concerning the analyses) and they present their results in a visually pleasing and intuitive way.

Weaknesses:

The authors used data from four different strip cropping experiments and there is no real replication in space as all of these differed in many aspects (different crops, different areas between years, different combinations, design of the strip cropping (orientation and width), sampling effort and sample sizes of beetles (differing more than 35 fold between sites; L 100f); for more differences see L 237ff). The reader gets the impression that the authors stitched data from various places together that were not made to fit together. This may not be a problem per se but it surely limits the strength of the data as results for various crops may only be based on small samples from one or two sites (it is generally unclear how many samples were used for each crop/crop combination).

One of my major concerns is that it is completely unclear where carabids were collected. As some strips were 3m wide, some others were 6m and the monoculture plots large, it can be expected that carabids were collected at different distances from the plot edge. This alone, however, was conclusively shown to affect carabid assemblages dramatically and could easily outweigh the differences shown here if not accounted for in the models (see e.g. Boetzel et al. (2024) or Knapp et al. (2019) among many other studies on within field-distributions of carabids).

The authors hint at a related but somewhat different problem in L 137ff - carabid assemblages sampled in strips were sampled in closer proximity to each other than assemblages in monoculture fields which is very likely a problem. The authors did not check whether their results are spatially autocorrelated and this shortcoming is hard to account for as it would have required a much bigger, spatially replicated design in which distances are maintained from the beginning. This limitation needs to be stated more clearly in the manuscript.

Similarly, we know that carabid richness and density depend strongly on crop type (see e.g. Toivonen et al. (2022)) which could have biased results if the design is not balanced (this information is missing but it seems to be the case, see e.g. Celeriac in Almere in 2022).

A more basic problem is that the reader neither learns where traps were located, how missing traps were treated for analyses how many samples there were per crop or crop combination (in a simple way, not through Table S7 - there has to have been a logic in each of these field trials) or why there are differences in the number of samples from the same location and year (see Table S7). This information needs to be added to the methods section.

As carabid assemblages undergo rapid phenological changes across the year, assemblages that are collected at different phenological points within and across years cannot easily be compared. The authors would need to standardize for this and make sure that the assemblages they analyze are comparable prior to analyses. Otherwise, I see the possibility that the reported differences might simply be biased by phenology.

Surrounding landscape structure is known to affect carabid richness and density and could thus also bias observed differences between treatments at the same locations (lower overall richness => lower differences between treatments). Landscape structure has not been taken into account in any way.

In the statistical analyses, it is unclear whether the authors used estimated marginal means (as they should) - this needs to be clarified.

In addition, and as mentioned by Dr. Rasmann in the previous round (comment 1), the manuscript, in its current form, still suffers from simplified generalizations that 'oversell' the impact of the study and should be avoided. The authors restricted their analyses to ground beetles and based their conclusions on a design with many 'heterogeneities' - they should not draw conclusions for farmland biodiversity but stick to their system and report what they found. Although I understand the authors have previously stated that this is 'not practically

feasible', the reason for this comment is simply to say that the authors should not oversell their findings.

<https://doi.org/10.7554/eLife.104762.1.sa1>

Reviewer #3 (Public review):

Summary:

In this paper, the authors made a sincere effort to show the effects of strip cropping, a technique of alternating crops in small strips of several meters wide, on ground beetle diversity. They state that strip cropping can be a useful tool for bending the curve of biodiversity loss in agricultural systems as strip cropping shows a relative increase in species diversity (i.e. abundance and species richness) of the ground beetle communities compared to monocultures. Moreover, strip cropping has the added advantage of not having to compromise on agricultural yields.

Strengths:

The article is well written; it has an easily readable tone of voice without too much jargon or overly complicated sentence structure. Moreover, as far as reviewing the models in depth without raw data and R scripts allows, the statistical work done by the authors looks good. They have well thought out how to handle heterogeneous, yet spatially and temporarily correlated field data. The models applied and the model checks performed are appropriate for the data at hand. Combining RDA and PCA axes together is a nice touch.

Weaknesses:

The evidence for strip cropping bringing added value for biodiversity is mixed at best. Yes, there is an increase in relative abundance and species richness at the field level, but it is not convincingly shown this difference is robust or can be linked to clear structural and hypothesised advantages of the strip cropping system. The same results could have been used to conclude that there are only very limited signs of real added value of strip cropping compared to monocultures.

There are a number of reasons for this:

- (1) Significant differences disappear at crop level, as the authors themselves clearly acknowledge, meaning that there are no differences between pairs of similar crops in the strip cropping fields and their respective monoculture. This would mean the strips effectively function as "mini-monocultures". The significant relative differences at the field level could be an artifact of aggregation instead of structural differences between strip cropping and monocultures; with enough data points things tend to get significant despite large variance. This should have been elaborated further upon by the authors with additional analyses, designed to find out where differences originate and what it tells about the functioning of the system. Or it should have provided ample reason for cautioning in drawing conclusions about the supposed effectiveness of strip cropping based on these findings.
- (2) The authors report percentages calculated as relative change of species richness and abundance in strip cropping compared to monocultures after rarefaction. This is in itself correct, however, it can be rather tricky to interpret because the perspective on actual species richness and abundance in the fields and treatments is completely lost; the reported percentages are dimensionless. The authors could have provided the average cumulative number of species and abundance after rarefaction. Also, range and/or standard error would have been useful to provide information as to the scale of differences between treatments.

This could provide a new perspective on the magnitude of differences between the two treatments which a dimensionless percentage cannot.

(3) The authors appear to not have modelled the abundance of any of the dominant ground beetle species themselves. Therefore it becomes impossible to assess which important species are responsible (if any) for the differences found in activity density between stripcropping and monocultures and the possible life history traits related reasons for the differences, or lack thereof, that are found. A big advantage of using ground beetles is that many life history traits are well studied and these should be used whenever there is reason, as there clearly is in this case. Moreover, it is unclear which species are responsible for the difference in species richness found at the field level. Are these dominant species or singletons? Do the strip cropping fields contain species that are absent in the monoculture fields and are not the cause of random variation or sampling? Unfortunately, the authors do not report on any of these details of the communities that were found, which makes the results much less robust.

(4) In the discussion they conclude that there is only a limited amount of interstrip movement by ground beetles. Otherwise, the results of the crop-level statistical tests would have shown significant deviation from corresponding monocultures. This is a clear indication that the strips function more like mini-monocultures instead of being more than the sum of its parts.

(5) The RDA results show a modelled variable of differences in community composition between strip cropping and monoculture. Percentages of explained variation of the first RDA axis are extremely low, and even then, the effect of location and/or year appear to peak through (Figure S3), even though these are not part of the modelling. Moreover, there is no indication of clustering of strip cropping on the RDA axis, or in fact on the first principal component axis in the larger RDA models. This means the explanatory power of different treatments is also extremely low. The crop level RDA's show some clustering, but hardly any consistent pattern in either communities of crops or species correlations, indicating that differences between strip cropping and monocultures are very small.

Furthermore, there are a number of additional weaknesses in the paper that should be addressed:

The introduction lacks focus on the issues at hand. Too much space is taken up by facts on insect decline and land sharing vs. land sparing and not enough attention is spent on the scientific discussion underlying the statements made about crop diversification as a restoration strategy. They are simply stated as facts or as hypotheses with many references that are not mentioned or linked to in the text. An explicit link to the results found in the large number of references should be provided.

The mechanistic understanding of strip cropping is what is at stake here. Does strip cropping behave similarly to intercropping, a technique that has been proven to be beneficial to biodiversity because of added effects due to increased resource efficiency and greater plant species richness? This should be the main testing point and agenda of strip cropping. Do the biodiversity benefits that have been shown for intercropping also work in strip cropping fields? The ground beetles are one way to test this. Hypotheses should originate from this and should be stated clearly and mechanistically.

One could question how useful indicator species analysis (ISA) is for a study in which predominantly highly eurytopic species are found. These are by definition uncritical of their habitat. Is there any mechanistic hypothesis underlying a suspected difference to be found in preferences for either strip cropping or monocultures of the species that were expected to be caught? In other words, did the authors have any *a priori* reasons to suspect differences, or has this been an exploratory exercise from which unexplained significant results should be used with great caution?

However, setting these objections aside there are in fact significant results with strong species associations both with monocultures and strip cropping. Unfortunately, the authors do not dig deeper into the patterns found a posteriori either. Why would some species associate so strongly with strip cropping? Do these species show a pattern of pitfall catches that deviate from other species, in that they are found in a wide range of strips with different crops in one strip cropping field and therefore may benefit from an increased abundance of food or shelter? Also, why would so many species associate with monocultures? Is this in any way logical? Could it be an artifact of the data instead of a meaningful pattern? Unfortunately, the authors do not progress along these lines in the methods and discussion at all.

A second question raised in the introduction is whether the arable fields that form part of this study contain rare species. Unfortunately, the authors do not elaborate further on this. Do they expect rare species to be more prevalent in the strip cropping fields? Why? Has it been shown elsewhere that intercropping provides room for additional rare species?

Considering the implications the results of this research can have on the wider discussion of bending the curve and the effects of agroecological measures, bold claims should be made with extreme restraint and be based on extensive proof and robust findings. I am not convinced by the evidence provided in this article that the claim made by the authors that strip cropping is a useful tool for bending the curve of biodiversity loss is warranted.

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Author response:

We thank all reviewers for the highly detailed review and the time and effort which has been invested in this review. We have read their perspectives, questions and suggested improvements with great interest. We have reflected on the public review in detail and have made the first provisional responses which are outlined below. First, we would like to respond to four main issues pointed out by the editor and reviewers:

(1) *Lack of yield data in the manuscript:* There have been yield data collected in most of the sites and years of our study, and these have already been published and cited in our manuscript. In the appendix of our manuscript, we included a table with yield data for the sites and years in which the beetle diversity was studied. These data show that strip cropping does not cause a systematic yield reduction.

(2) *Sampling design clarification:* Our paper combines data from trials conducted at different locations and years. On the one hand this allows an analysis of a comprehensive dataset, but on the other hand in some cases there were slight inconsistencies in how data were collected or processed (e.g. taxonomic level of species identification). We will explain the sampling design and data analysis in more detail to increase clarity and transparency.

(3) *Additional data analysis:* In the revised manuscript we will present an analysis on the responses of abundances of the 12 most common ground beetle genera to strip cropping. This will give better insight of the variation in responses among ground beetle taxa.

(4) *Restrict findings to our system:* We will nuance our findings further and will focus more strongly on the implications of our data on ground beetle communities, rather than on agrobiodiversity in a broader sense.

We will further work on improving the manuscript based on reviewers feedback in the coming weeks, aiming to submit a revised version of the manuscript at the end of February.

Detailed response to editor and reviewers:

Editor Comments:

(1) You only have analyzed ground beetle diversity, it would be important to add data on crop yields, which certainly must be available (note that in normal intercropping these would likely be enhanced as well).

Most yield data have been published in three previous papers, which we already cited or will cite (one was not yet published at the time of submission). Our argumentation is based on these studies. We had also already included a table in the appendix that showed the yield data that relates specifically to our locations and years of measurement. The finding that strip cropping does not majorly affect yield is based on these findings. We will consider changing the title of our manuscript to remove the explicit focus on yield.

(2) Considering the heterogeneous data involving different experiments it is particularly important to describe the sampling design in detail and explain how various hierarchical levels were accounted for in the analysis.

We agree that some important details to our analysis were not described in sufficient detail. Especially reviewer 2 pointed out several relevant points that we did account for in our analyses, but which were not clear from the text in the methods section. We are convinced that our data analyses are robust and that our conclusions are supported by the data. We will revise the methods section to make our approach clearer and more transparent.

(3) In addition to relative changes in richness and density of ground beetles you should also present the data from which these have been derived. Furthermore, you could also analyze and interpret the response of the different individual taxa to strip cropping.

With our heterogeneous dataset it was quite complicated to show overall patterns of absolute changes in ground beetle abundance and richness, especially for the field-level analyses. As the sampling design was not always the same and occasionally samples were missing, the number of year series that made up a datapoint were different among locations and years. However, we always made sure that for the comparison of a paired monoculture and strip cropping field, the number of year series was always made equal through rarefaction. That is, the number of ground beetle(s) (species) are always expressed as the number per 2 to 6 samples. Therefore, we prefer to stick to relative changes as we are convinced that this gives a fairer representation of our complex dataset.

We agree with the second point that both the editor and several reviewers pointed out. The indicator species analyses that we used were biased by rare species, and we now omit this analysis. Instead, we will include a GLM analysis on the responses of abundances of the 12 most common ground beetle genera to strip cropping. We chose for genera here (and not species) as we could then include all locations and years within the analysis, and in most cases a genus was dominated by a single species (but notable exceptions were *Amara* and *Harpalus*, which were made up of several species). We will illustrate these findings still in a similar fashion as we did for the indicator species analysis.

(4) Keep to your findings and don't overstate them but try to better connect them to basic ecological hypotheses potentially explaining them.

After careful consideration of the important points that reviewers point out, we decided to nuance our points about biodiversity conservation along two key lines: (1) the extent to which ground beetles can be indicators of wider biodiversity changes; and (2) our findings

that are not as straightforward positive as our narrative suggests. We still believe that strip cropping contributes positively to carabid communities, and will carefully check the text to avoid overstatements.

Reviewer 1:

Summary:

This study demonstrates that strip cropping enhances the taxonomic diversity of ground beetles across organically-managed crop systems in the Netherlands. In particular, strip cropping supported 15% more ground beetle species and 30% more individuals compared to monocultures.

Strengths:

A well-written study with well-analyzed data of a complex design. The data could have been analyzed differently e.g. by not pooling samples, but there are pros and cons for each type of analysis and I am convinced this will not affect the main findings. A strong point is that data were collected for 4 years. This is especially strong as most data on biodiversity in cropping systems are only collected for one or two seasons. Another strong point is that several crops were included.

We thank reviewer 1 for their kind words and agree with this strength of the paper. The paper combines data from trials conducted at different locations and years. On the one hand this allows an analysis of a comprehensive dataset, but on the other hand in some cases there were slight inconsistencies in how data were collected or processed (e.g. taxonomic level of species identification).

Weaknesses:

This study focused on the biodiversity of ground beetles and did not examine crop productivity. Therefore, I disagree with the claim that this study demonstrates biodiversity enhancement without compromising yield. The authors should present results on yield or, at the very least, provide a stronger justification for this statement.

We acknowledge that we indeed did not formally analyze yield in our study, but we have good reason for this. The claim that strip cropping does not compromise yield comes from several extensive studies (Juventia et al., 2024; Ditzler et al., 2023; Carillo-Reche et al., 2023) that were conducted in nearly all the sites and years that we included in our study. We chose not to include formal analyses of productivity for two key reasons: (1) a yield analysis would duplicate already published analyses, and (2) we prefer to focus more on the ecology of ground beetles and the effect of strip cropping on biodiversity, rather than diverging our focus also towards crop productivity. Nevertheless, we have shown the results on yield in Table S6 and refer extensively to the studies that have previously analyzed this data.

Reviewer 2:

Summary:

The authors aimed to investigate the effects of organic strip cropping on carabid richness and density as well as on crop yields. They find on average higher carabid richness and density in strip cropping and organic farming, but not in all cases.

Strengths:

Based on highly resolved species-level carabid data, the authors present estimates for many different crop types, some of them rarely studied, at the same time. The authors

did a great job investigating different aspects of the assemblages (although some questions remain concerning the analyses) and they present their results in a visually pleasing and intuitive way.

We appreciate the kind words of reviewer 2 and their acknowledgement of the extensiveness of our dataset. In our opinion, the inclusion of many different crops is indeed a strength, rarely seen in similar studies; and we are happy that the figures are appreciated.

Weaknesses:

The authors used data from four different strip cropping experiments and there is no real replication in space as all of these differed in many aspects (different crops, different areas between years, different combinations, design of the strip cropping (orientation and width), sampling effort and sample sizes of beetles (differing more than 35 fold between sites; L 100f); for more differences see L 237ff). The reader gets the impression that the authors stitched data from various places together that were not made to fit together. This may not be a problem per se but it surely limits the strength of the data as results for various crops may only be based on small samples from one or two sites (it is generally unclear how many samples were used for each crop/crop combination).

The paper indeed combines data from trials conducted at different locations and years. On the one hand this allows an analysis of a comprehensive dataset, but on the other hand in some cases there were slight differences in the experimental design. At the time that we did our research, there were only a handful of farmers that were employing strip cropping within the Netherlands, which greatly reduced the number of fields for our study. Therefore, we worked in the sites that were available and studied as many crops on these sites. Since there was variation in the crops grown in the sites, for some crops we have limited replication. In the revision we will explain this more clearly.

One of my major concerns is that it is completely unclear where carabids were collected. As some strips were 3m wide, some others were 6m and the monoculture plots large, it can be expected that carabids were collected at different distances from the plot edge. This alone, however, was conclusively shown to affect carabid assemblages dramatically and could easily outweigh the differences shown here if not accounted for in the models (see e.g. Boetzel et al. (2024) or Knapp et al. (2019) among many other studies on within field-distributions of carabids).

Point well taken and we will present a more detailed description of the sampling design in the methods. Samples were always taken at least 10 meters into the field, and always in the middle of the strip. This would indeed mean that there is a small difference between the 3- and 6m wide strips regarding distance from another strip, but this was then only a difference of 1.5 to 3 meters from the edge. A difference that, based on our own extensive experience with ground beetle communities, will not have a large impact on the findings of ground beetles. The distance from field/plot edges was similar between monocultures and strip cropped fields.

The authors hint at a related but somewhat different problem in L 137ff - carabid assemblages sampled in strips were sampled in closer proximity to each other than assemblages in monoculture fields which is very likely a problem. The authors did not check whether their results are spatially autocorrelated and this shortcoming is hard to account for as it would have required a much bigger, spatially replicated design in which distances are maintained from the beginning. This limitation needs to be stated more clearly in the manuscript.

This is a limitation that is hard to avoid in comparisons between strip cropping and monoculture systems because the use of a statistically robust design with sufficient replication and still using field sizes that are representative for farming practice are often not possible. We will acknowledge this limitation in the revised manuscript. To allow a fair comparison based on sufficient number of replications, we chose to combine data from several years and locations (despite this not being the ideal experimental design). This approach has the drawback that ground beetle communities are difficult to compare. Therefore, we chose to further investigate two years of data from Wageningen as the factorial design allowed a fair comparison between monocultures and strip cropping. We analyzed three crop combinations during two years, but we still cannot exclude a potential influence of spatial autocorrelation. We acknowledged this limitation in our original submission, and we will clarify this point further in the revision.

Similarly, we know that carabid richness and density depend strongly on crop type (see e.g. Toivonen et al. (2022)) which could have biased results if the design is not balanced (this information is missing but it seems to be the case, see e.g. Celeriac in Almere in 2022).

The samples size ranges between 2 and 6 per combination of cropping design, crop, location and year. We believe that this will allow a meaningful analysis. Moreover, our main focus is the comparison between monoculture and strip cropping, and not the comparison between different crops. Even though we show that crop types have different ground beetle communities, we are most interested in the contrast of ground beetle communities in strip cropping and monoculture systems.

A more basic problem is that the reader neither learns where traps were located, how missing traps were treated for analyses how many samples there were per crop or crop combination (in a simple way, not through Table S7 - there has to have been a logic in each of these field trials) or why there are differences in the number of samples from the same location and year (see Table S7). This information needs to be added to the methods section.

Point well taken. We will clarify this further in the revised manuscript. As we combined data from several experimental designs that originally had slightly different research questions, this in part caused differences between numbers of rounds or samples per crop, location or year.

As carabid assemblages undergo rapid phenological changes across the year, assemblages that are collected at different phenological points within and across years cannot easily be compared. The authors would need to standardize for this and make sure that the assemblages they analyze are comparable prior to analyses. Otherwise, I see the possibility that the reported differences might simply be biased by phenology.

We agree and we dealt with this issue by using year series instead of using individual samples of different rounds. While this approach is not perfect, it allows us to get the best possible impression of the entire ground beetle community across seasons. For our analyses we had the choice to only include data from sampling rounds that were conducted at the same time, or to include all available data. We chose to analyze all data, and made sure that the number of samples between strip cropping and monoculture fields per location, year and crop was always the same by pooling and rarefaction. In this way we have analyzed a complex multi-year, multi-crop and multi-location dataset as good as we could.

Surrounding landscape structure is known to affect carabid richness and density and could thus also bias observed differences between treatments at the same locations (lower overall richness => lower differences between treatments). Landscape structure has not been taken into account in any way.

We did not include landscape structure as there are only 4 sites, which does not allow a meaningful analysis of potential effects landscape structure. Studying how landscape interacts with strip cropping to influence insect biodiversity would require at least, say 15 to 20 sites, which was not feasible for this study. However, such an analysis may be possible in an ongoing project (CropMix) which includes many farms that work with strip cropping.

In the statistical analyses, it is unclear whether the authors used estimated marginal means (as they should) - this needs to be clarified.

In the revised manuscript we will further clarify this point.

In addition, and as mentioned by Dr. Rasmann in the previous round (comment 1), the manuscript, in its current form, still suffers from simplified generalizations that 'oversell' the impact of the study and should be avoided. The authors restricted their analyses to ground beetles and based their conclusions on a design with many 'heterogeneities' - they should not draw conclusions for farmland biodiversity but stick to their system and report what they found. Although I understand the authors have previously stated that this is 'not practically feasible', the reason for this comment is simply to say that the authors should not oversell their findings.

In the revised manuscript, we will nuance our findings by explaining that strip cropping is a potentially useful tool to support ground beetle biodiversity in agricultural fields, but the effects on other taxa still needs to be further explored.

Reviewer 3:

Summary:

In this paper, the authors made a sincere effort to show the effects of strip cropping, a technique of alternating crops in small strips of several meters wide, on ground beetle diversity. They state that strip cropping can be a useful tool for bending the curve of biodiversity loss in agricultural systems as strip cropping shows a relative increase in species diversity (i.e. abundance and species richness) of the ground beetle communities compared to monocultures. Moreover, strip cropping has the added advantage of not having to compromise on agricultural yields.

Strengths:

The article is well written; it has an easily readable tone of voice without too much jargon or overly complicated sentence structure. Moreover, as far as reviewing the models in depth without raw data and R scripts allows, the statistical work done by the authors looks good. They have well thought out how to handle heterogeneous, yet spatially and temporarily correlated field data. The models applied and the model checks performed are appropriate for the data at hand. Combining RDA and PCA axes together is a nice touch.

We thank reviewer 3 for their kind words and appreciation for the simple language and analysis that we used.

Weaknesses:

The evidence for strip cropping bringing added value for biodiversity is mixed at best. Yes, there is an increase in relative abundance and species richness at the field level, but it is not convincingly shown this difference is robust or can be linked to clear structural and hypothesised advantages of the strip cropping system. The same results could have been used to conclude that there are only very limited signs of real added value of strip cropping compared to monocultures.

Point well taken. We agree that the effect of strip cropping on carabid beetle communities are subtle and we will nuance the text in the revised version to reflect this.

There are a number of reasons for this:

(1) Significant differences disappear at crop level, as the authors themselves clearly acknowledge, meaning that there are no differences between pairs of similar crops in the strip cropping fields and their respective monoculture. This would mean the strips effectively function as "mini-monocultures".

This is indeed in line with our conclusions. Based on our data and results, the advantages of strip cropping seem mostly to occur because crops with different communities are now on a same field, rather than that within the strips you get mixtures of communities related to different crops. We discussed this in the first paragraph of the discussion in the original submission.

The significant relative differences at the field level could be an artifact of aggregation instead of structural differences between strip cropping and monocultures; with enough data points things tend to get significant despite large variance. This should have been elaborated further upon by the authors with additional analyses, designed to find out where differences originate and what it tells about the functioning of the system. Or it should have provided ample reason for cautioning in drawing conclusions about the supposed effectiveness of strip cropping based on these findings.

We believe that this is a misunderstanding of our approach. In the field-level analyses we pooled samples from the same field (i.e. pseudo-replicates were pooled), resulting in a relatively small sample size of 50 samples. We will explain this better in the methods section. Therefore, the statement “with enough data points things tend to get significant” is not applicable here.

(2) The authors report percentages calculated as relative change of species richness and abundance in strip cropping compared to monocultures after rarefaction. This is in itself correct, however, it can be rather tricky to interpret because the perspective on actual species richness and abundance in the fields and treatments is completely lost; the reported percentages are dimensionless. The authors could have provided the average cumulative number of species and abundance after rarefaction. Also, range and/or standard error would have been useful to provide information as to the scale of differences between treatments. This could provide a new perspective on the magnitude of differences between the two treatments which a dimensionless percentage cannot.

We agree that this would be the preferred approach if we would have had a perfectly balanced dataset. However, this approach is not feasible with our unbalanced design and differences in sampling effort. While we acknowledge the limitation of the interpretation of percentages, it does allow reporting relative changes for each combination of location, year and crop. The number of samples on which the percentages were based were always kept

equal (through rarefaction) between the cropping systems (for each combination of location, year and crop), but not among crops, years and location. The reason for this is that we did not always have an equal number of samples available between both cropping systems, and this approach allowed us to make a better estimation whenever more samples were available. For example, sometimes we had 2 samples from a strip cropped field and 6 from the monoculture, here we would use rarefaction up to 2 samples (where we would just have a better estimation from the monoculture). In other cases, we had 4 samples in both strip cropped and monoculture field, here we chose to use rarefaction to 4 samples to get a better estimation altogether. Adding a value for actual richness or abundance to the figures would have distorted these findings, as the variation would be huge (as it would represent the number of ground beetle(s) species per 2 to 6 pitfall samples). Furthermore, the dimension that reviewer 3 describes would thus be “The number of ground beetle species / individuals per 2 to 6 samples”, not a very informative unit either. We chose to trade-off better estimations of difference between cropping systems over a more readily interpretable unit.

(3) The authors appear to not have modelled the abundance of any of the dominant ground beetle species themselves. Therefore it becomes impossible to assess which important species are responsible (if any) for the differences found in activity density between strip cropping and monocultures and the possible life history traits related reasons for the differences, or lack thereof, that are found. A big advantage of using ground beetles is that many life history traits are well studied and these should be used whenever there is reason, as there clearly is in this case. Moreover, it is unclear which species are responsible for the difference in species richness found at the field level. Are these dominant species or singletons? Do the strip cropping fields contain species that are absent in the monoculture fields and are not the cause of random variation or sampling? Unfortunately, the authors do not report on any of these details of the communities that were found, which makes the results much less robust.

Thank you for raising this point. We have reconsidered our indicator species analysis and found that it is rather sensitive for rare species and insensitive for changes in common species. Therefore, we will replace the indicator species analyses with a GLM analysis for the 12 most common genera of ground beetles in the revised manuscript. This will allow us to go more in depth on specific traits of the genera which abundances change depending on the cropping system. In the revised manuscript, we will also discuss these common genera more in depth, rather than focusing on rarer species. Furthermore, we will add information on rarity and habitat preference to the table that shows species abundances per location (Table S2).

(4) In the discussion they conclude that there is only a limited amount of interstrip movement by ground beetles. Otherwise, the results of the crop-level statistical tests would have shown significant deviation from corresponding monocultures. This is a clear indication that the strips function more like mini-monocultures instead of being more than the sum of its parts.

This is in line with our point in the first paragraph of the discussion and an important message of our manuscript.

(5) The RDA results show a modelled variable of differences in community composition between strip cropping and monoculture. Percentages of explained variation of the first RDA axis are extremely low, and even then, the effect of location and/or year appear to peak through (Figure S3), even though these are not part of the modelling. Moreover, there is no indication of clustering of strip cropping on the RDA axis, or in fact on the first principal component axis in the larger RDA models. This means the explanatory power of different treatments is also extremely low. The crop level RDA's show some clustering, but

hardly any consistent pattern in either communities of crops or species correlations, indicating that differences between strip cropping and monocultures are very small.

We agree and we make a similar point in the first paragraph of the discussion.

Furthermore, there are a number of additional weaknesses in the paper that should be addressed:

The introduction lacks focus on the issues at hand. Too much space is taken up by facts on insect decline and land sharing vs. land sparing and not enough attention is spent on the scientific discussion underlying the statements made about crop diversification as a restoration strategy. They are simply stated as facts or as hypotheses with many references that are not mentioned or linked to in the text. An explicit link to the results found in the large number of references should be provided.

We will streamline the introduction by omitting the land sharing vs. land sparing topic and better linking references to our research findings.

The mechanistic understanding of strip cropping is what is at stake here. Does strip cropping behave similarly to intercropping, a technique that has been proven to be beneficial to biodiversity because of added effects due to increased resource efficiency and greater plant species richness? This should be the main testing point and agenda of strip cropping. Do the biodiversity benefits that have been shown for intercropping also work in strip cropping fields? The ground beetles are one way to test this. Hypotheses should originate from this and should be stated clearly and mechanistically.

We agree with the reviewer and will clarify this research direction clearer in the introduction of the revised manuscript.

One could question how useful indicator species analysis (ISA) is for a study in which predominantly highly eurytopic species are found. These are by definition uncritical of their habitat. Is there any mechanistic hypothesis underlying a suspected difference to be found in preferences for either strip cropping or monocultures of the species that were expected to be caught? In other words, did the authors have any a priori reasons to suspect differences, or has this been an exploratory exercise from which unexplained significant results should be used with great caution?

Point well taken. We agree that the indicator species analysis has limitations and therefore now replaced this with GLM analysis for the 12 most common ground beetle genera.

However, setting these objections aside there are in fact significant results with strong species associations both with monocultures and strip cropping. Unfortunately, the authors do not dig deeper into the patterns found a posteriori either. Why would some species associate so strongly with strip cropping? Do these species show a pattern of pitfall catches that deviate from other species, in that they are found in a wide range of strips with different crops in one strip cropping field and therefore may benefit from an increased abundance of food or shelter? Also, why would so many species associate with monocultures? Is this in any way logical? Could it be an artifact of the data instead of a meaningful pattern? Unfortunately, the authors do not progress along these lines in the methods and discussion at all.

We thank reviewer 3 for these valuable perspectives. In the revised manuscript, we will further explore the species/genera that respond to cropping systems and discuss these findings in more detail.

A second question raised in the introduction is whether the arable fields that form part of this study contain rare species. Unfortunately, the authors do not elaborate further on this. Do they expect rare species to be more prevalent in the strip cropping fields? Why? Has it been shown elsewhere that intercropping provides room for additional rare species?

The answer is simply no, we did not find more rare species in strip cropping. In the revised manuscript, we will add a column for rarity (according to waarneming.nl) in the table showing abundances of species per location. We only found two rare species, one of which we only found a single individual and one that was more related to the open habitat created by a failed wheat field. We will discuss this more in depth in the discussion.

Considering the implications the results of this research can have on the wider discussion of bending the curve and the effects of agroecological measures, bold claims should be made with extreme restraint and be based on extensive proof and robust findings. I am not convinced by the evidence provided in this article that the claim made by the authors that strip cropping is a useful tool for bending the curve of biodiversity loss is warranted.

We believe that strip cropping can be a useful tool because farmers readily adopt it and it can result in modest biodiversity gains without yield loss. However, strip cropping is indeed not a silver bullet (which we also don't claim). We will nuance the implications of our study in the revised manuscript.

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